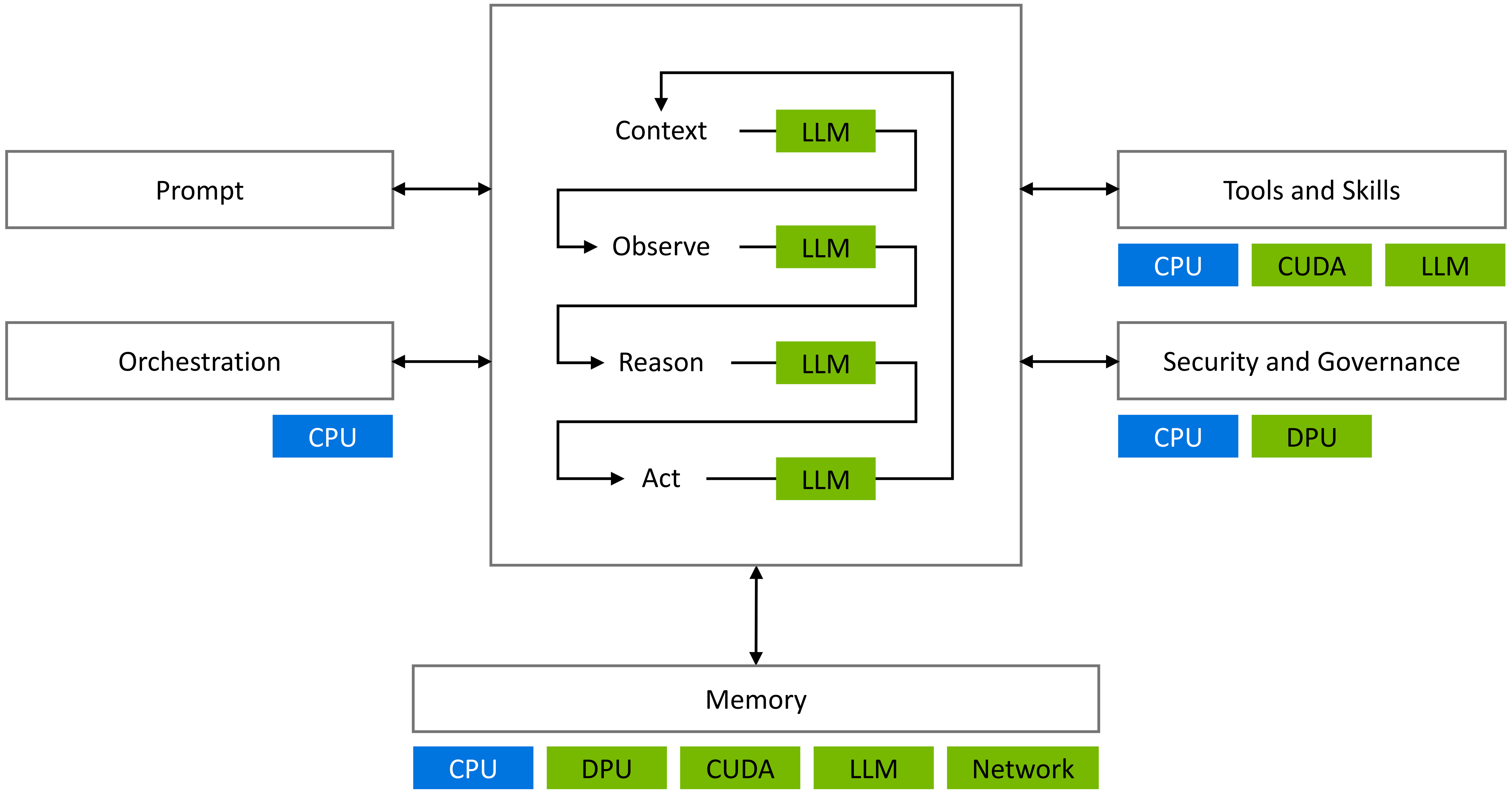
Think Fast: LPU Accelerator for Heterogeneous Compute

Igor Arsovski, Santosh Raghavan | NVIDIA

Hot Chips 2026

Agentic AI is the Most Complex Computing Workload in History



NVIDIA Vera Rubin is a Full-Stack AI Factory Platform

Extreme co-design for agentic AI across seven chips and five racks

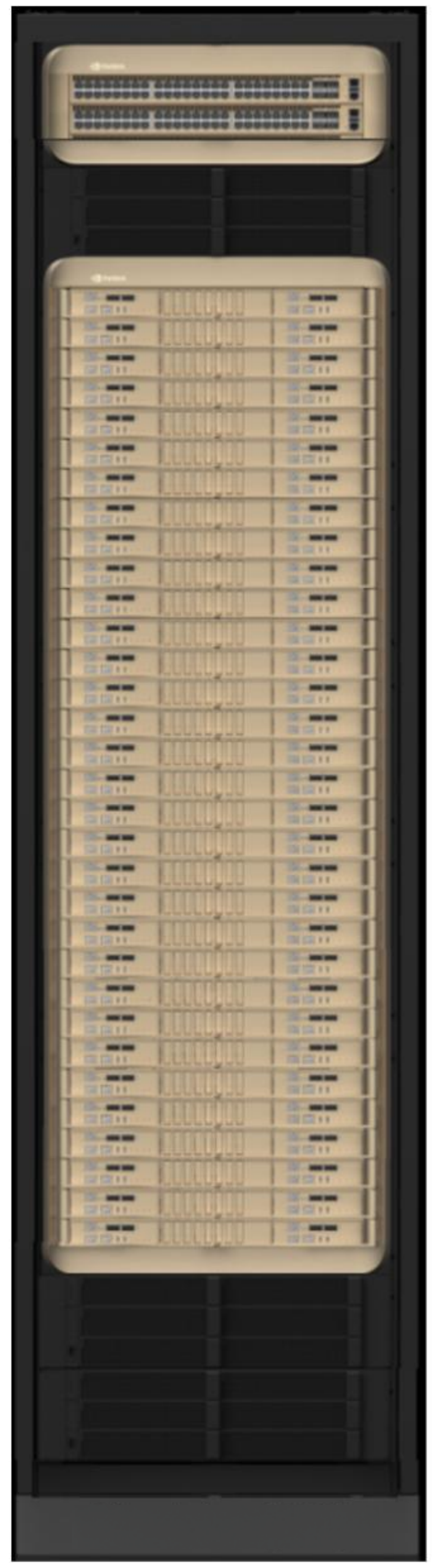
Vera Rubin NVL72

Foundation Platform

Groq 3 LPX

Extend Interactivity

Vera CPU Rack



Tool Calls and Sandboxes

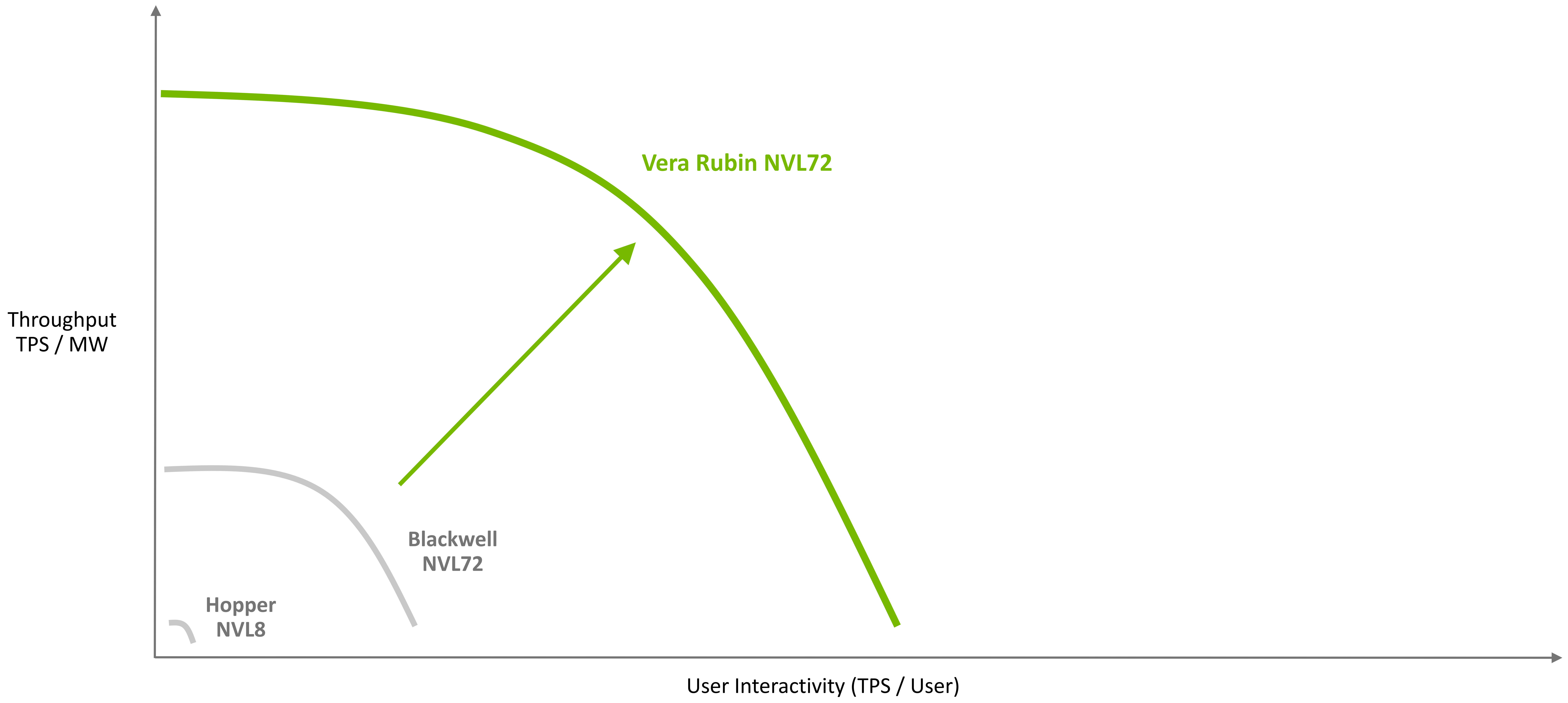
Vera BlueField-4 STX Storage

Context Memory Storage

Spectrum-6 SPX Networking

Scale-out Fabric

Vera Rubin NVL72 is the Foundation of Every AI Factory



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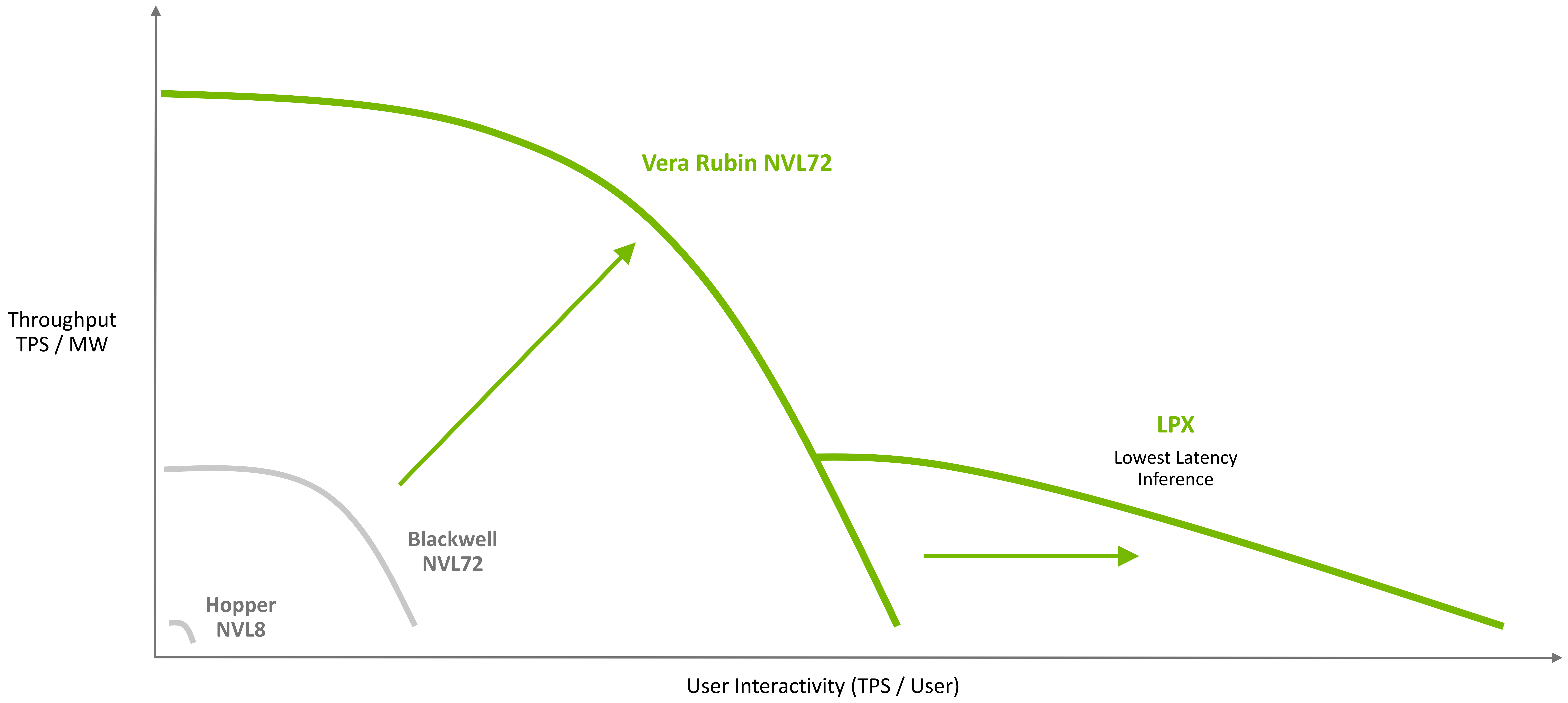
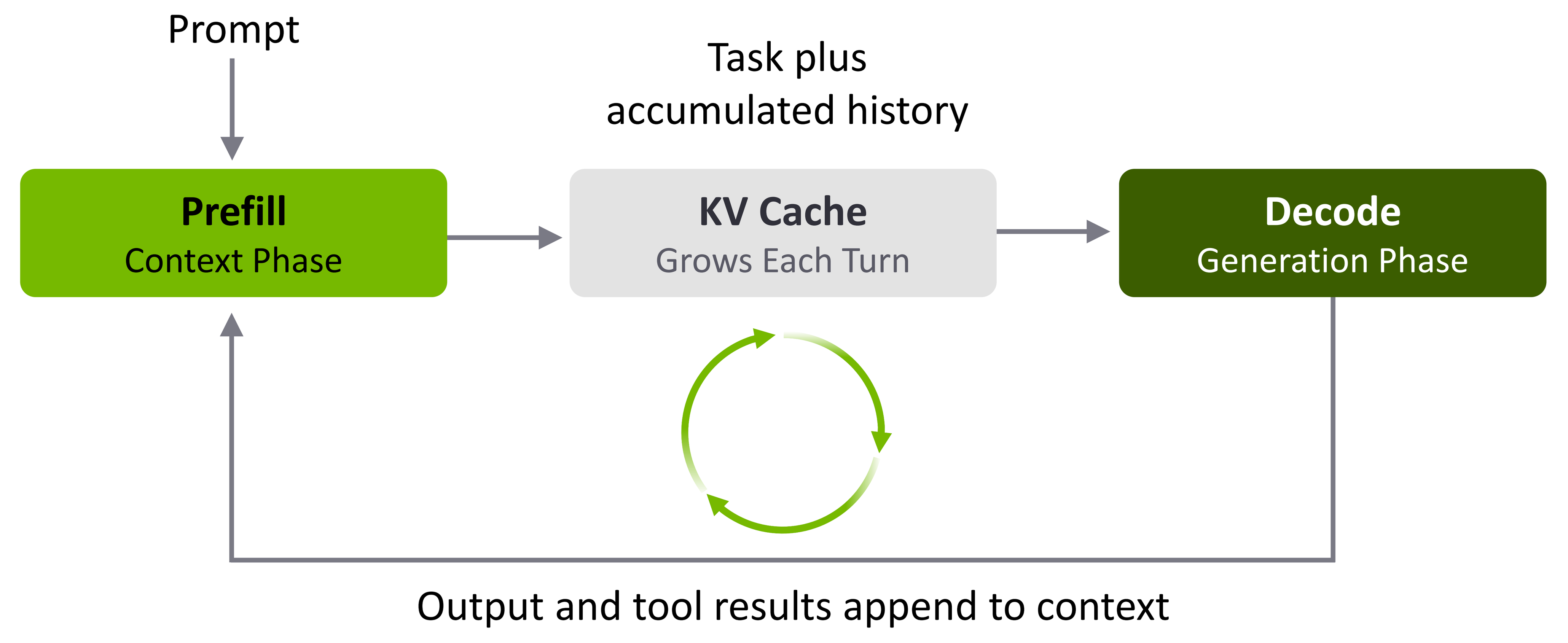
NVIDIA Groq 3 LPX — The Interactive AI Inference Accelerator

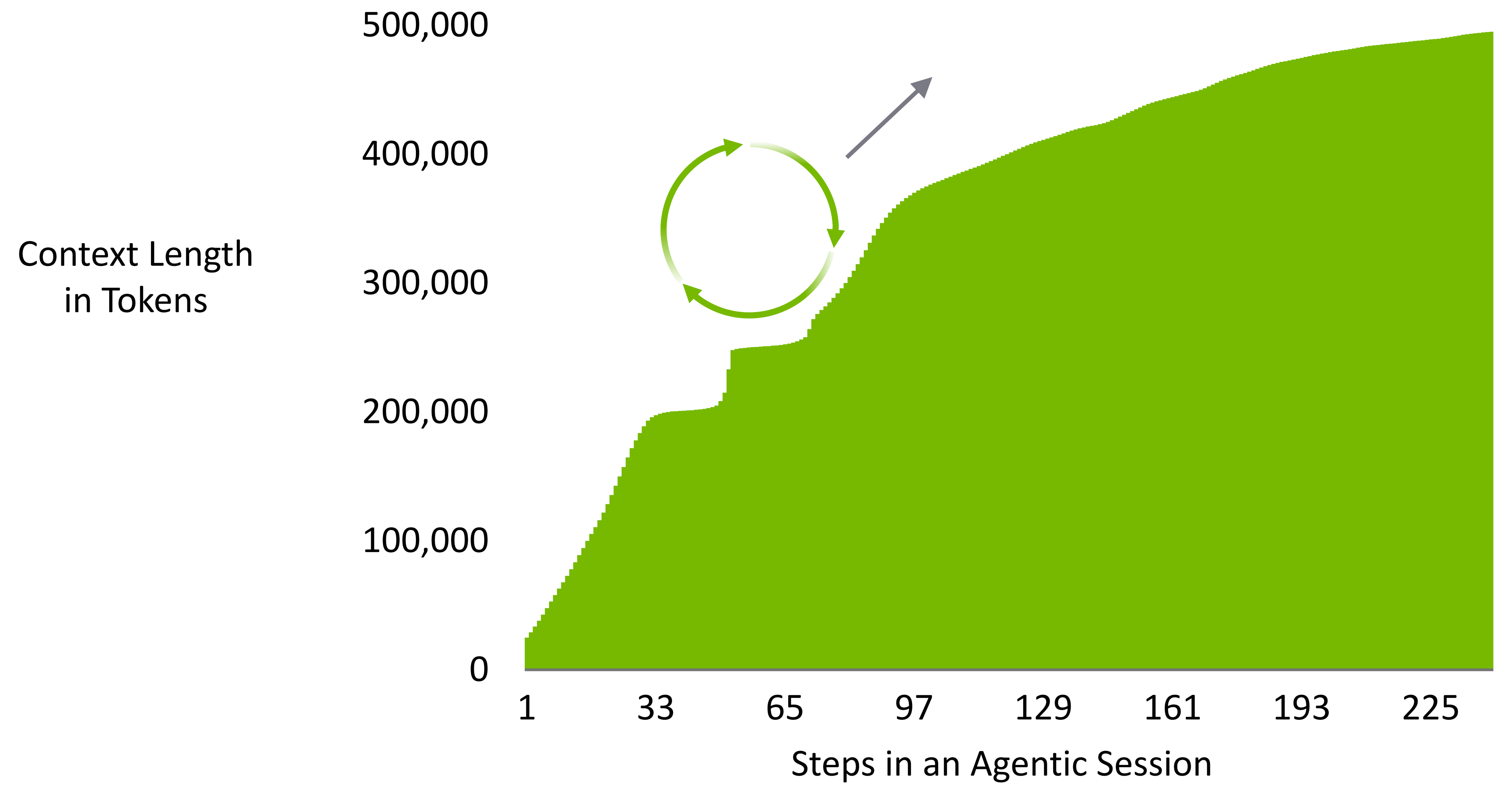
Ultra fast user experiences for agentic workloads

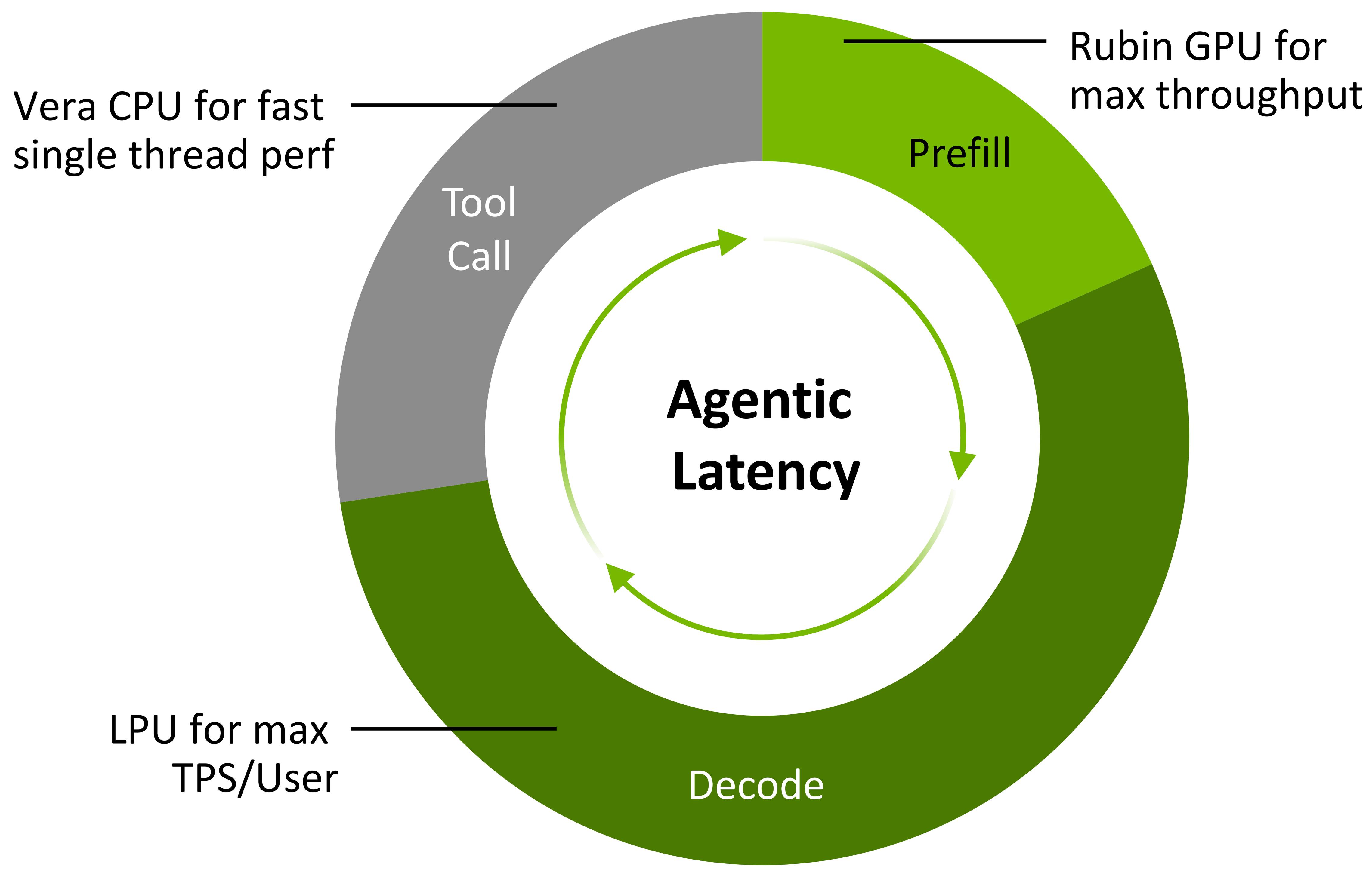


Agentic AI Compounds Context Tokens and Latency

Hundreds of agent steps increase context and slow user experience



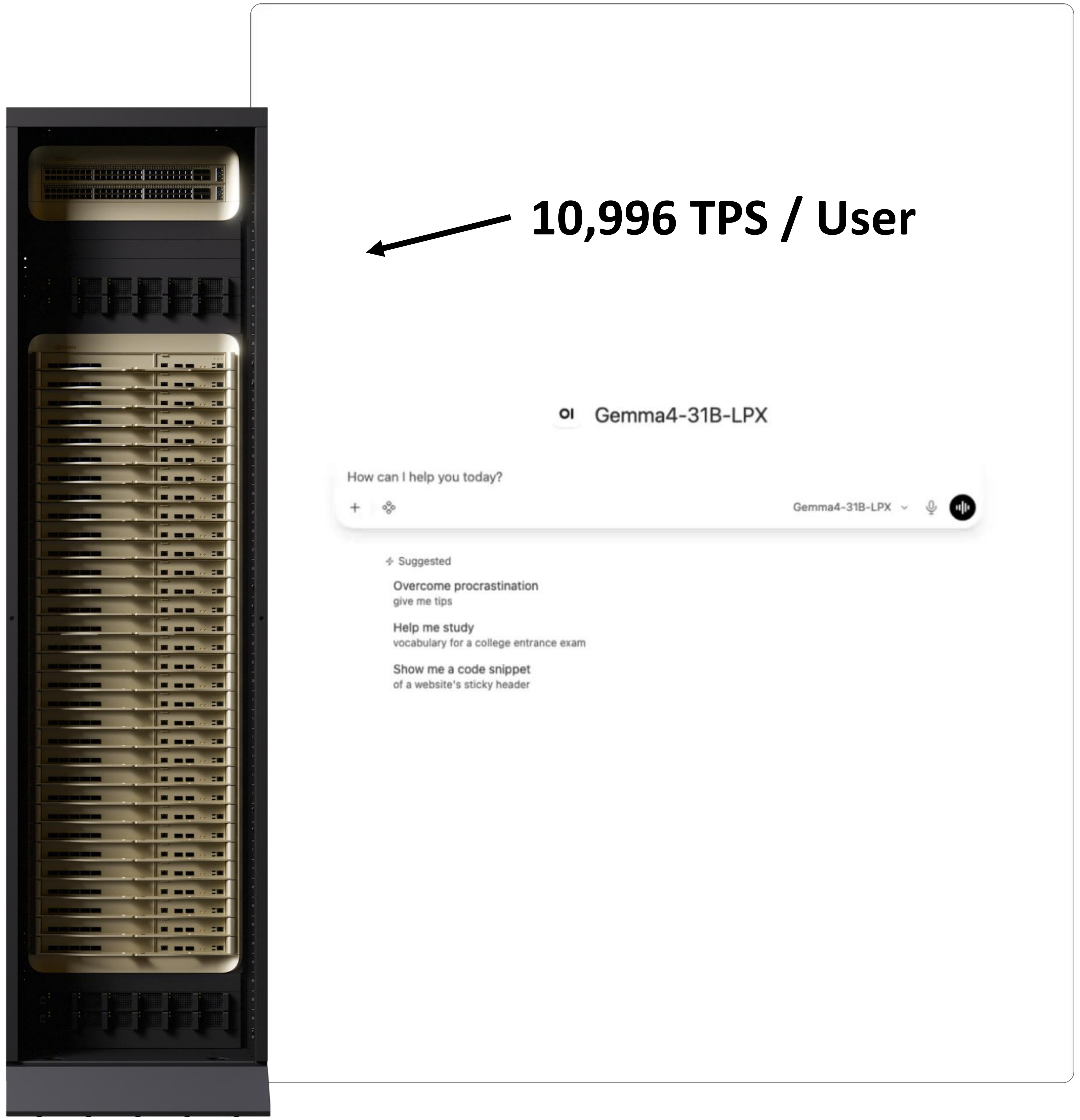




Groq 3 LPX – Engineered at the Speed of Light

11,000 Tokens Per Second Sets a New Bar for Inference AI

NVIDIA Groq 3 LPX rack

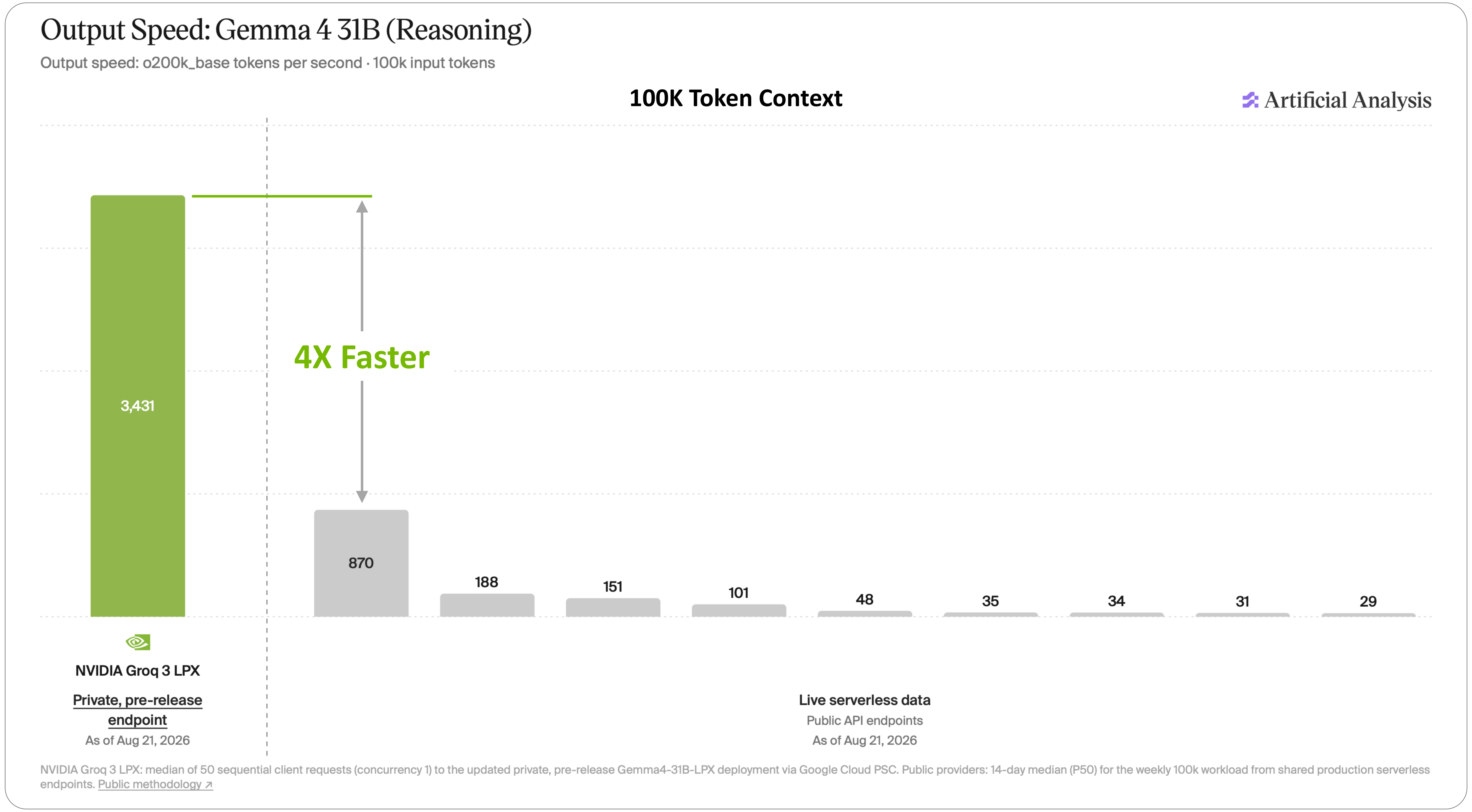


Example Coding Gemma 4
31B–16K ISL Out of Max
264K Context

SPEED-bench coding running at
4,767 median output tokens / second

LPX Delivers 4x Faster Long-Context Decode in 3rd Party Benchmarking

Groq 3 LPX Measured by Artificial Analysis on 100K context length for agents



NVIDIA Groq 3 LPU

Igor Arsovski

Near-SRAM Compute Eliminates Memory Bottleneck for Low-Latency Inference

VLIW Architecture with small instruction control area overhead

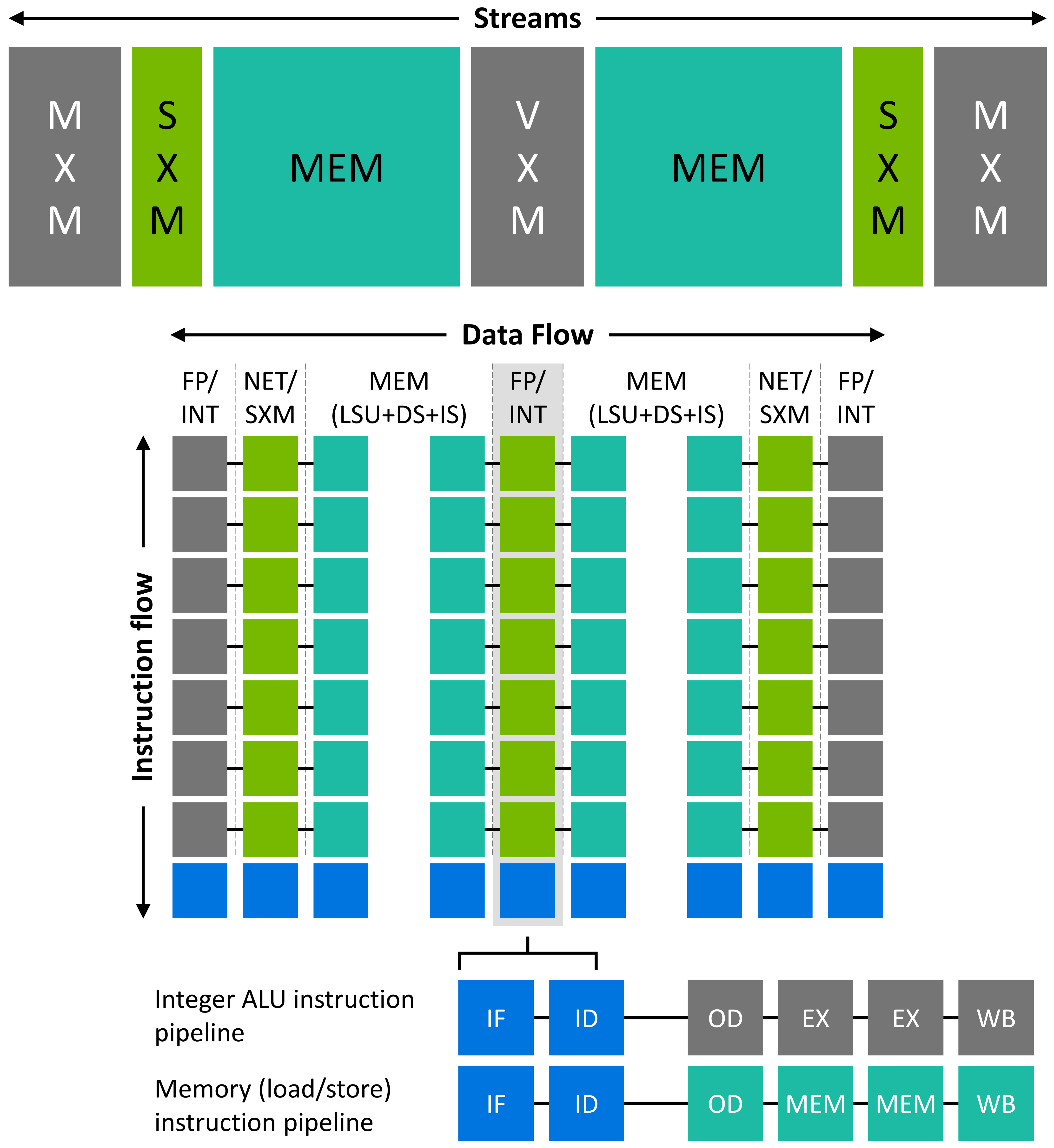
Flat memory hierarchy: entirely in SRAM

Compute fed via high-bandwidth “streams”

All units operate synchronously across multiple LPUs

Functional units:

- **MEM:** On-chip SRAM
- **VXM:** Vector PEs
- **MXM:** Matrix of MAC
- **SXM:** Data reshapes



Software-Scheduled Execution Makes Maximum Throughput Predictable

SW schedules processing at CLK-period granularity

Flat memory hierarchy: entirely in SRAM

Compute fed via high-bandwidth "streams"

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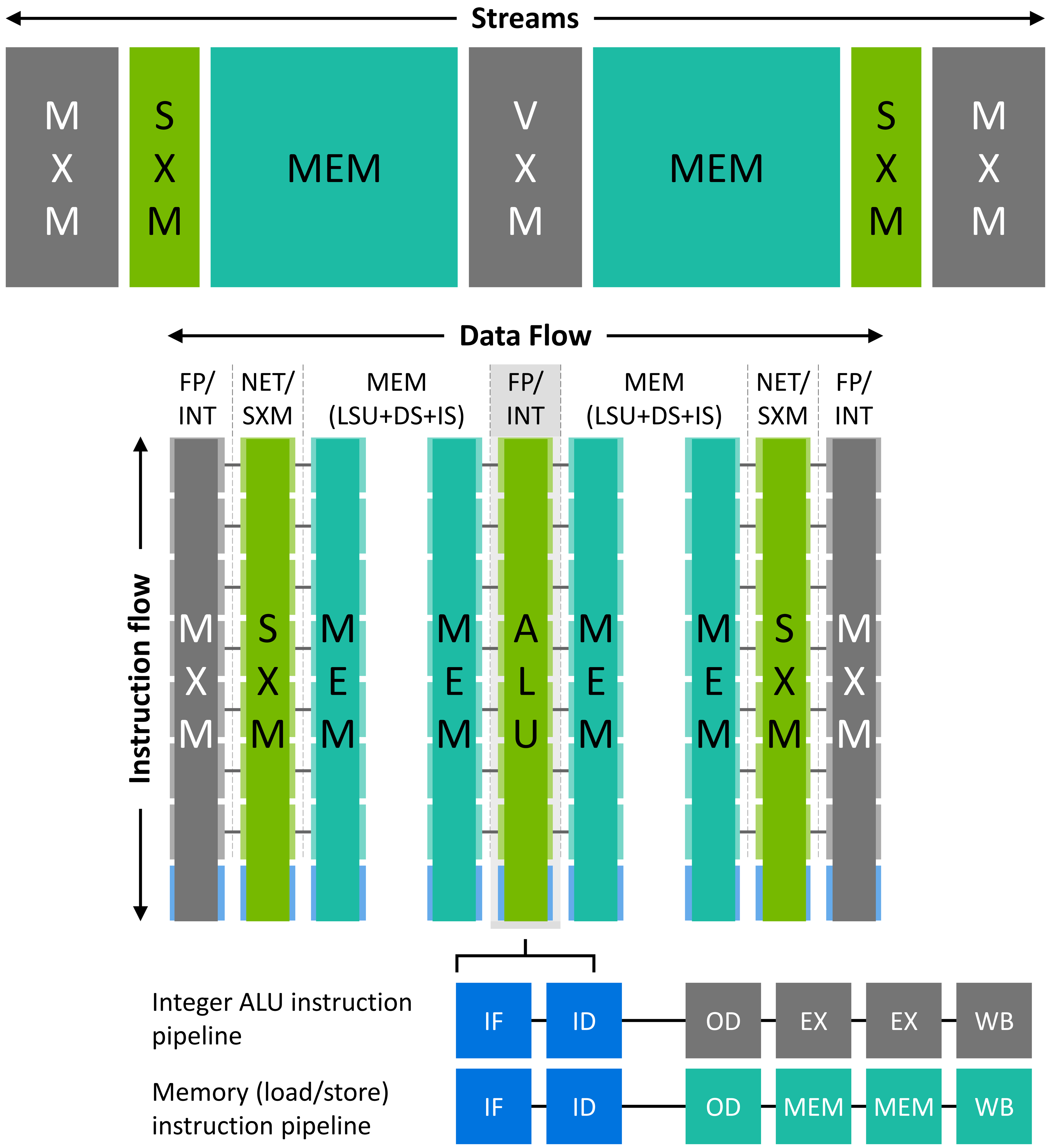
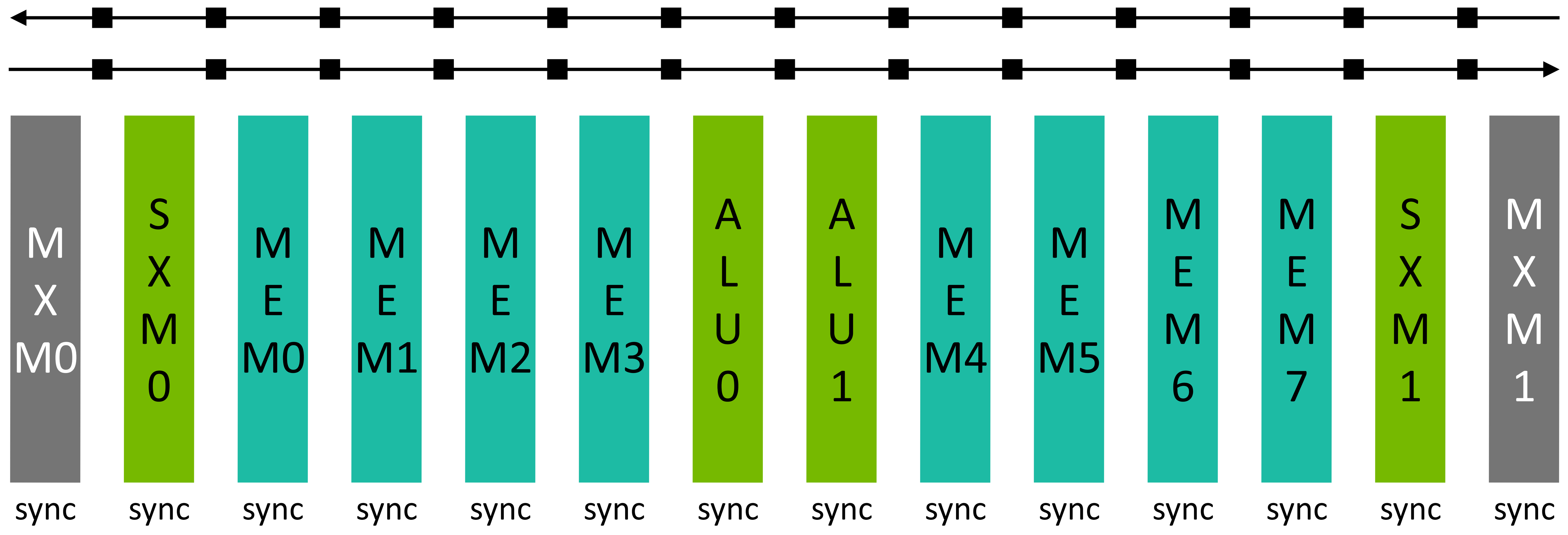


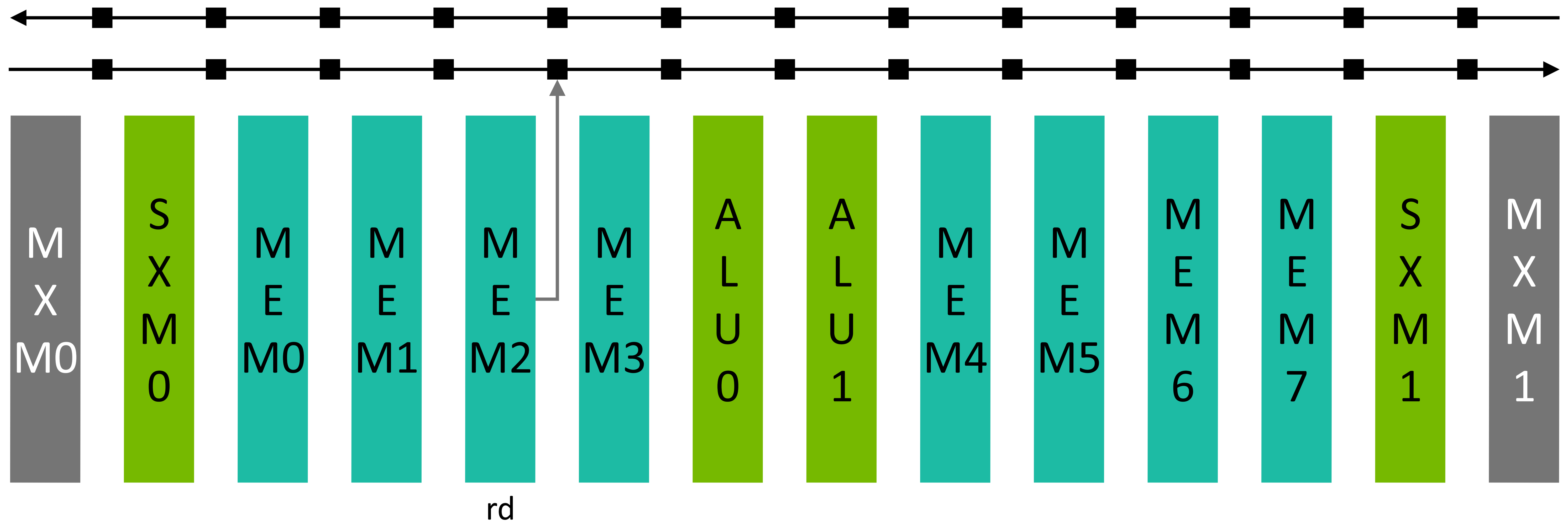
LPU Core: SIMD Units Synchronized for Lockstep Execution

All SIMD units synchronized to single global clock for lockstep instruction dispatch



LPU Core Simple Schedule Example

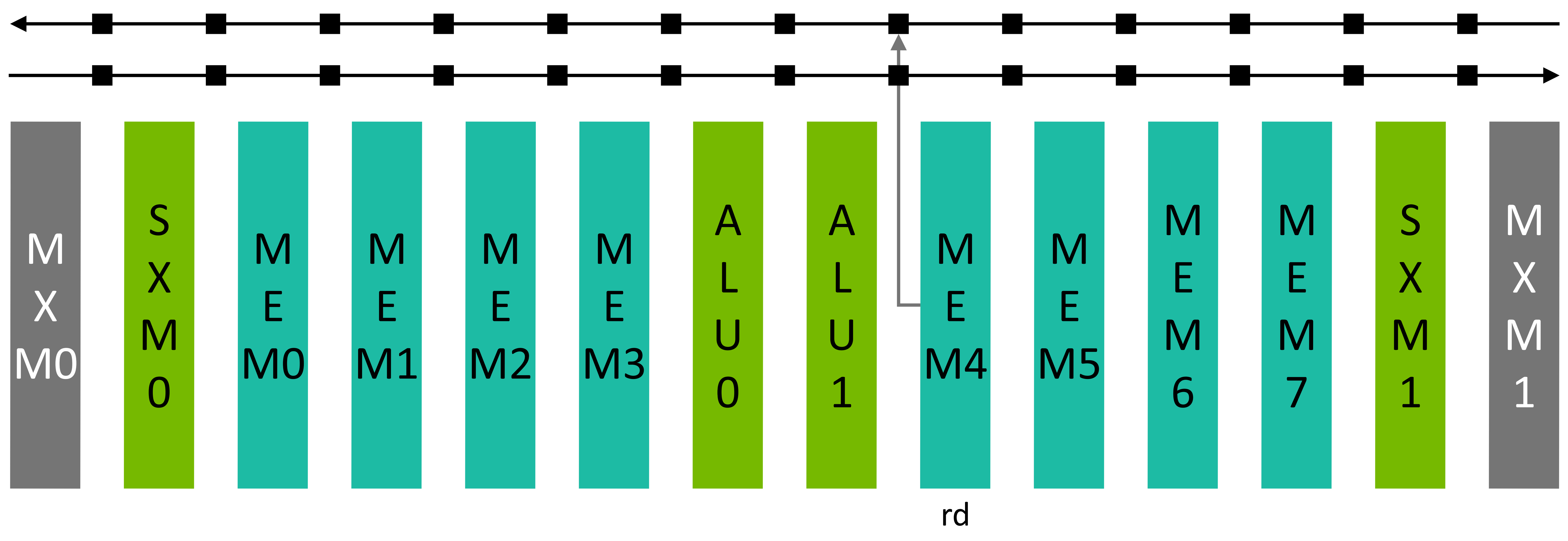
Read SRAM and place first operand on processor stream



Cycle	Event
0	rd MEM2

LPU Core Simple Schedule Example

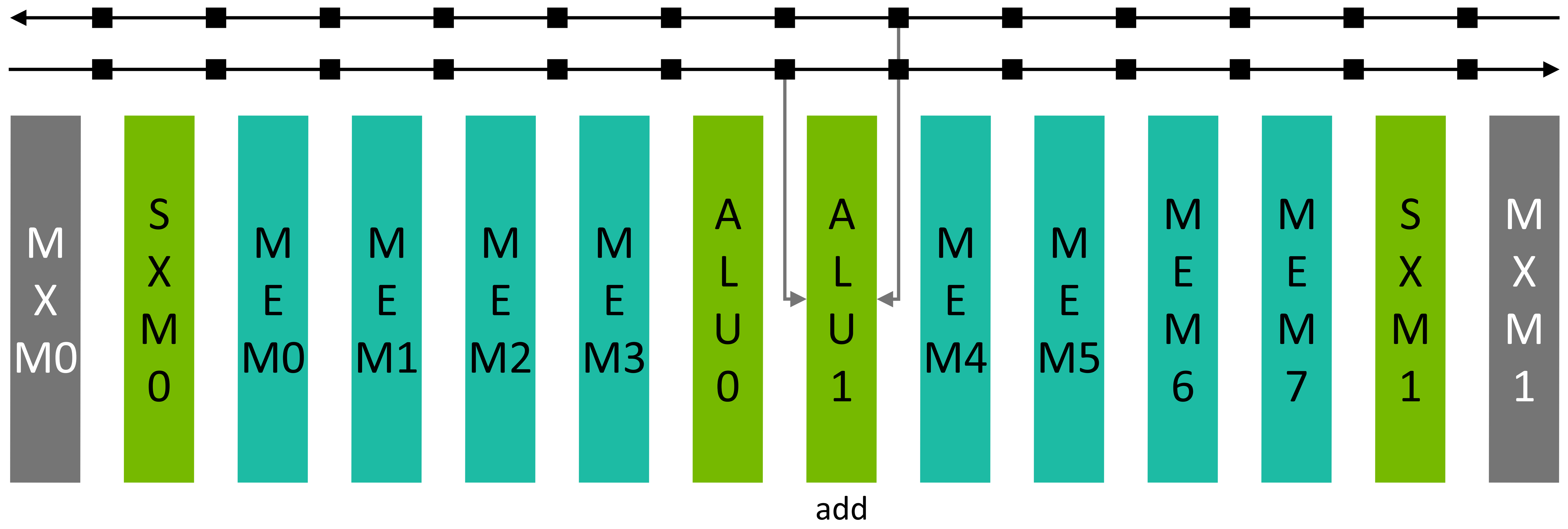
Read SRAM and place second operand on processor stream



Cycle	Event
0	rd MEM2
2	rd MEM 4

LPU Core Simple Schedule Example

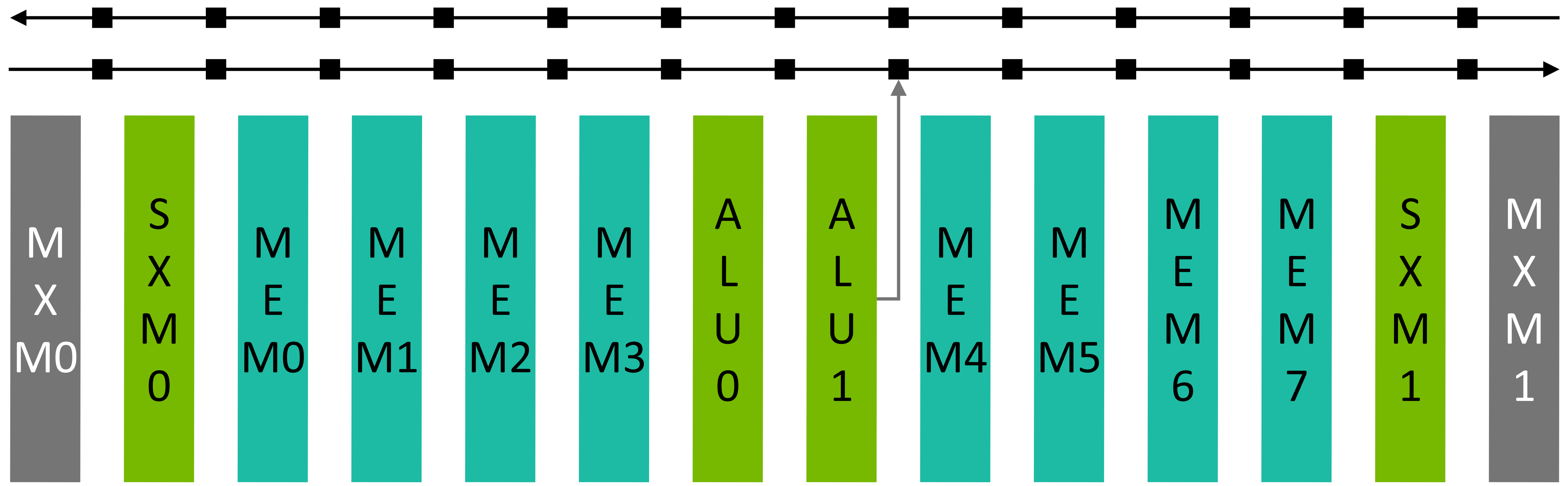
Perform add on first and second operand



Cycle	Event
0	rd MEM2
2	rd MEM 4
3	add ALU1

LPU Core Simple Schedule Example

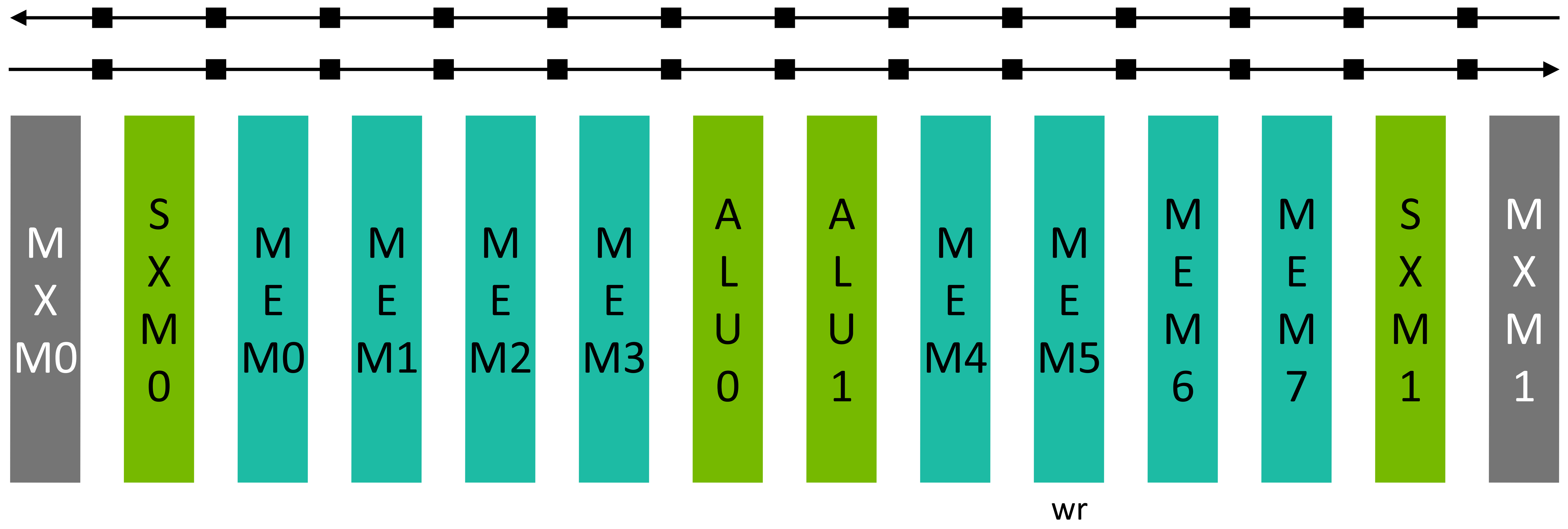
Place addition results on a stream



Cycle	Event
0	rd MEM2
2	rd MEM 4
3	add ALU1
5	*appear on stream*

LPU Core Simple Schedule Example

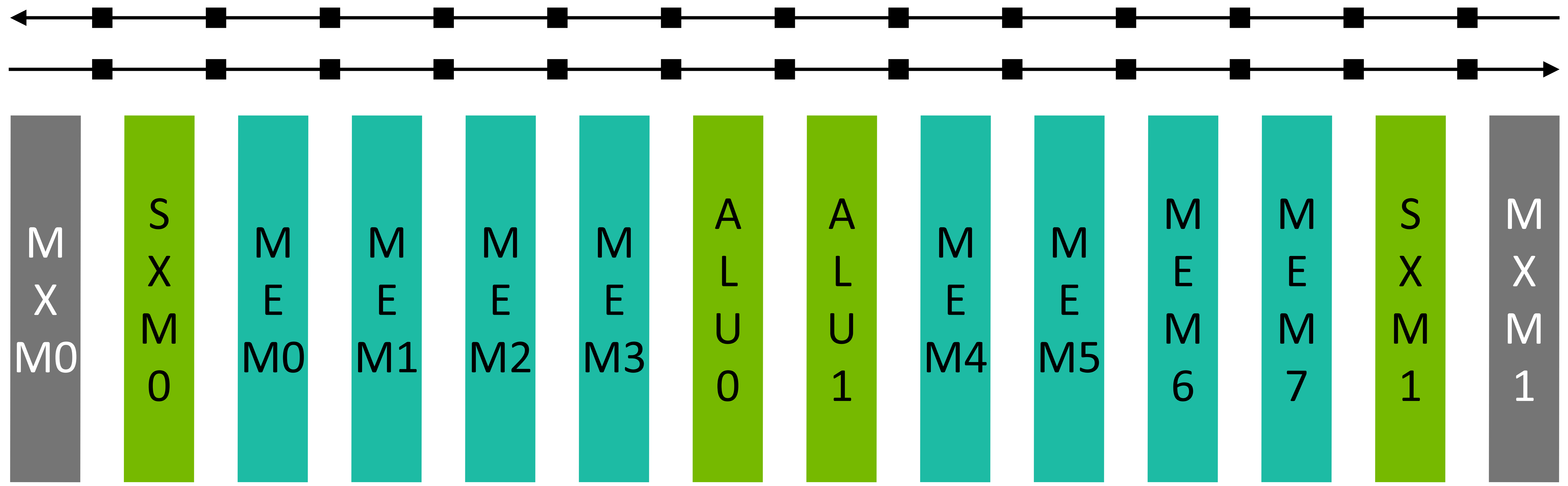
Write results in SRAM



Cycle	Event
0	rd MEM2
2	rd MEM 4
3	add ALU1
5	*appear on stream*
7	wr MEM5

LPU Core Simple Schedule Example

Multiple operations occur concurrently using high stream BW and large SIMD concurrency



	Concurrent Schedules	
Cycle	Event	Event
0	rd MEM2	rd MEM3
1		rd MEM4
2	rd MEM 4	add ALU1
3	add ALU1	
4		*appear on stream*
5	*appear on stream*	wr MEM4
7	wr MEM5	

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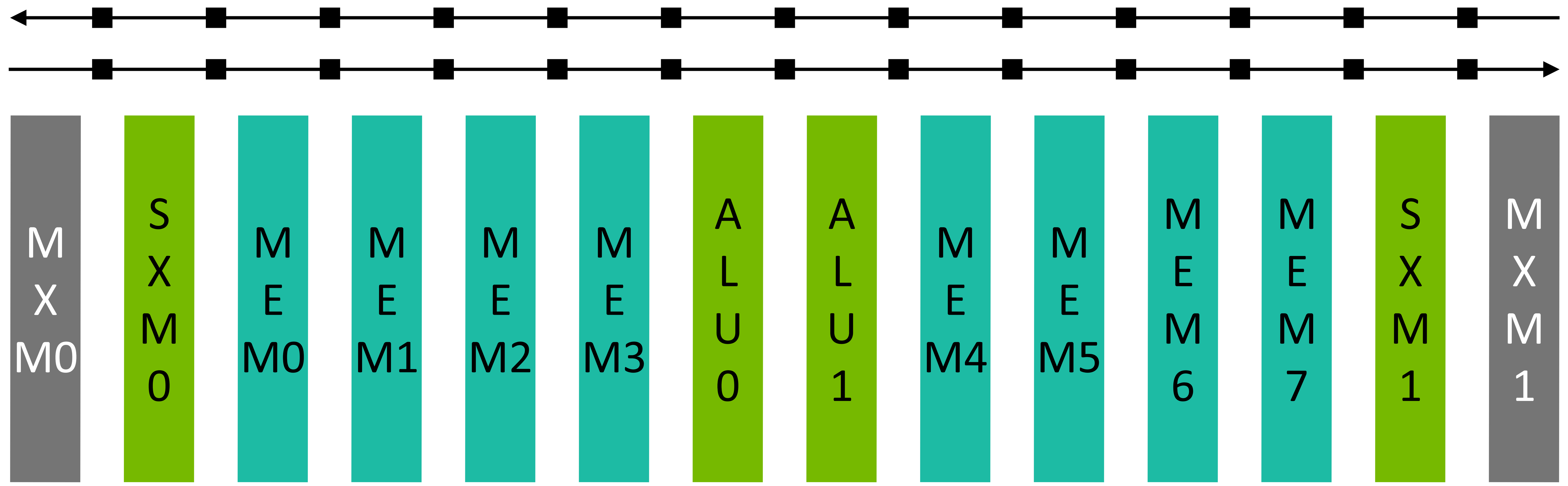
LPU Core Simple Schedule Example

No kernel boundaries | No MEMBARs | All operations “fused”



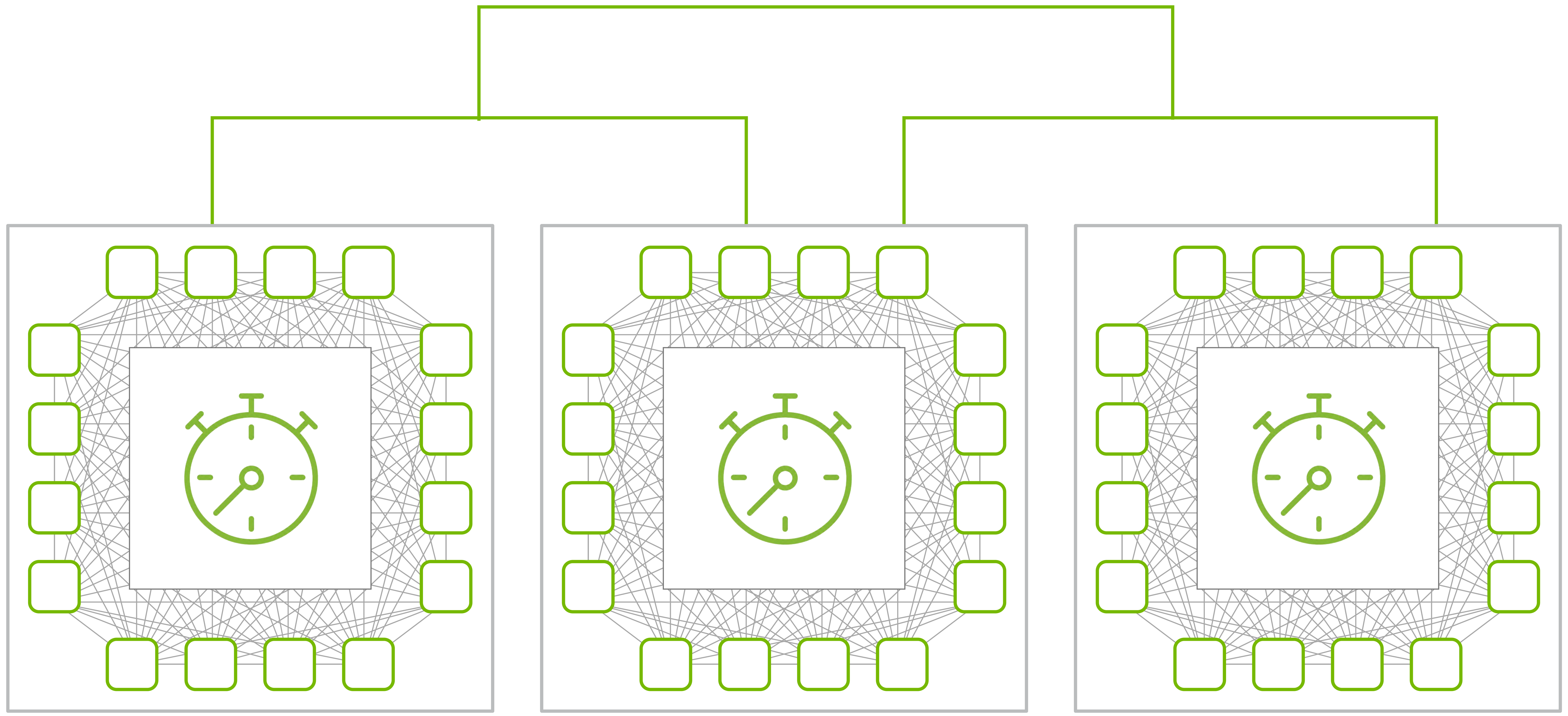
	Concurrent Schedules	
Cycle	Event	Event
0	rd MEM2	rd MEM3
1		rd MEM4
2	rd MEM 4	add ALU1
3	add ALU1	
4		*appear on stream*
5	*appear on stream*	wr MEM4
7	wr MEM5	
8	rd MEM5	
10	exp ALU1	

1000+ Chips Together Make a Massive Single LPU Core

LPU software-scheduled network built on a single virtual global clock

LPU network synchronizes all chips within and across racks to work on the same global clock acting like a single core cluster

Clock drift across chips is accounted for and mitigated deterministically



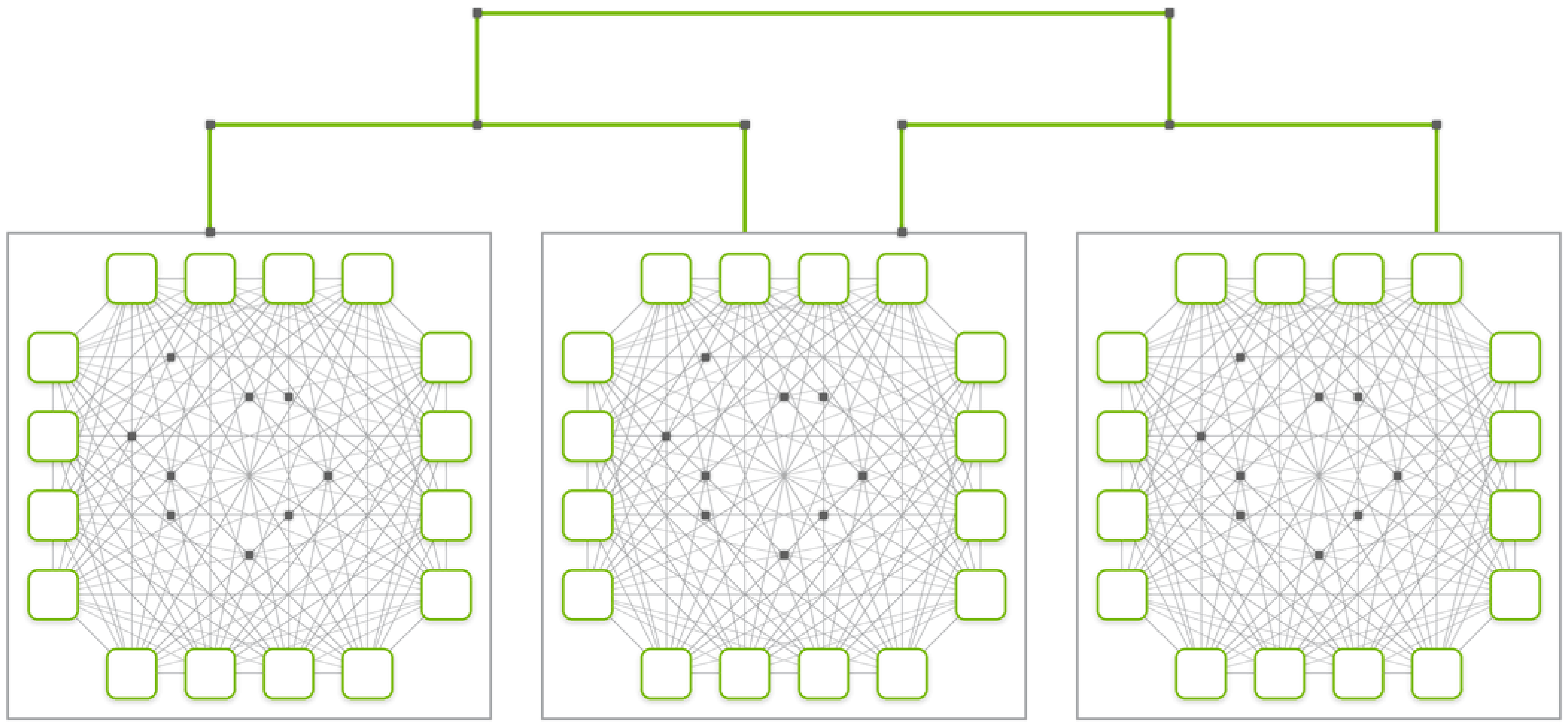
Low-Diameter, Low-Overhead Networking Keeps Communication From Becoming the Bottleneck

Combining processor and router in a single LPU

Each chip acts as both a processor and router

Compiler schedules messages as part
of programs loaded onto each chip

No adaptive routing / congestion sensing needed



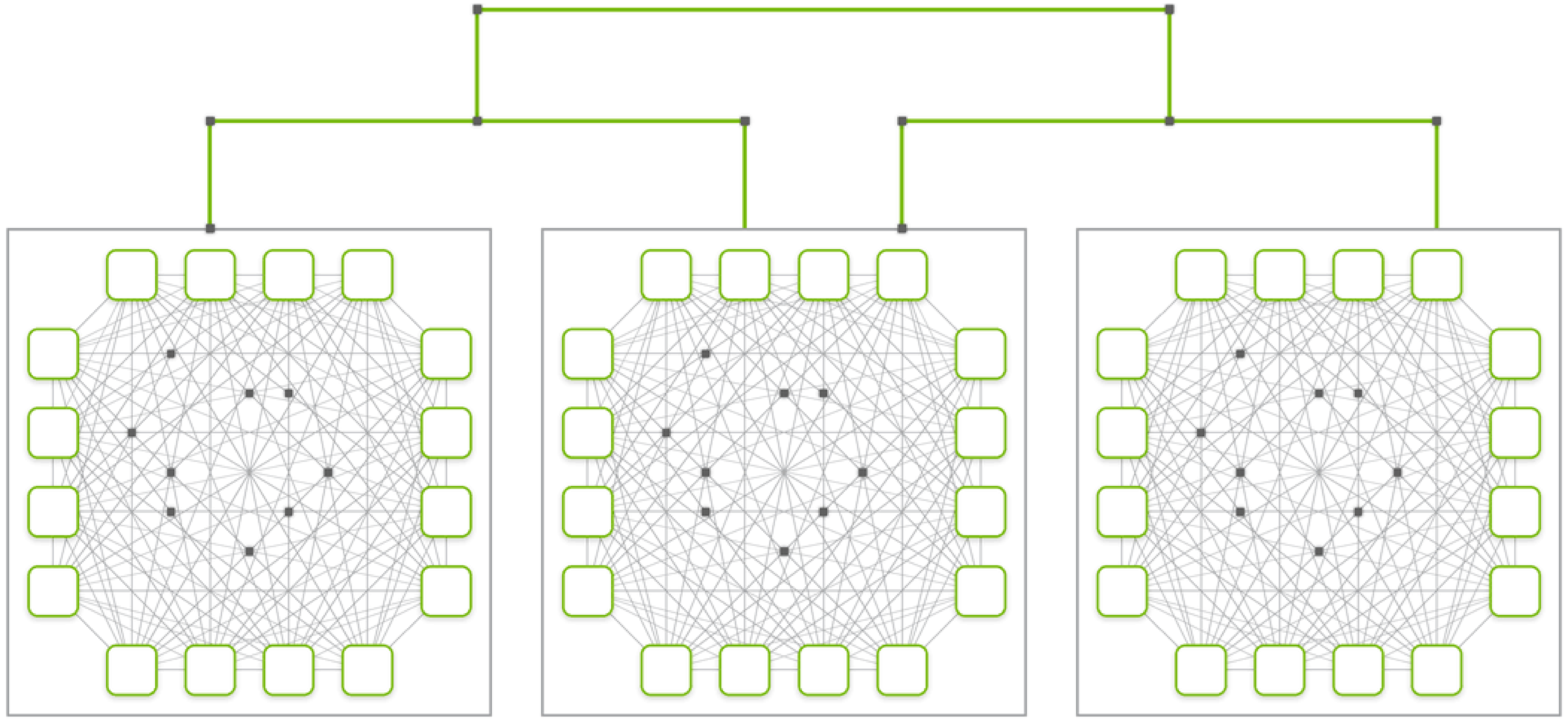
Low Packet Overhead Enables Fast Network Communication

Routing tensors, not packets

Low encoding overhead, from the header + tail flit

No hardware flow control, no virtual channel information, etc. (Eth. Layer-1)

Improves performance (bandwidth) especially at smaller message (tensor) sizes for inference



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Low Latency Scale-Up Interconnect Enables Fast Inference

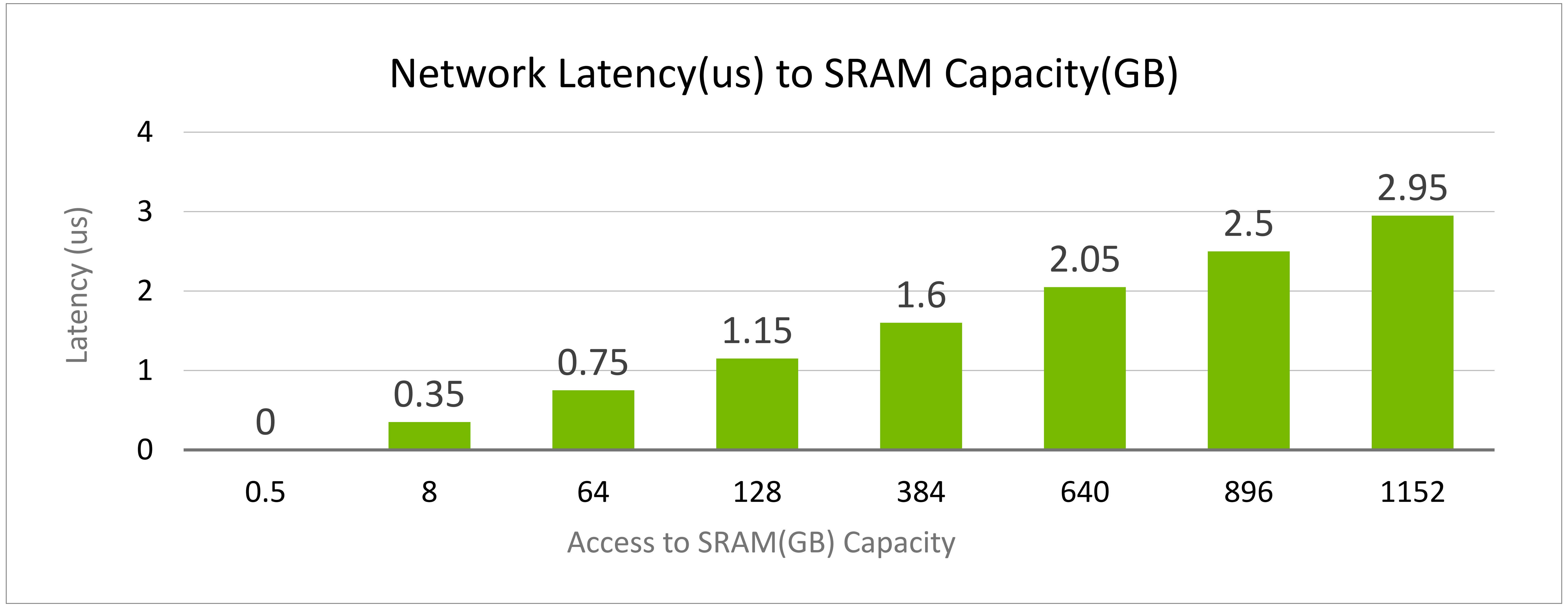
NVIDIA Groq 3 LPX Rack Overview

256 LPUs

128GB SRAM Capacity

315 PFLOPs FP8 fed with 40 PB/s of aggregate SRAM bandwidth

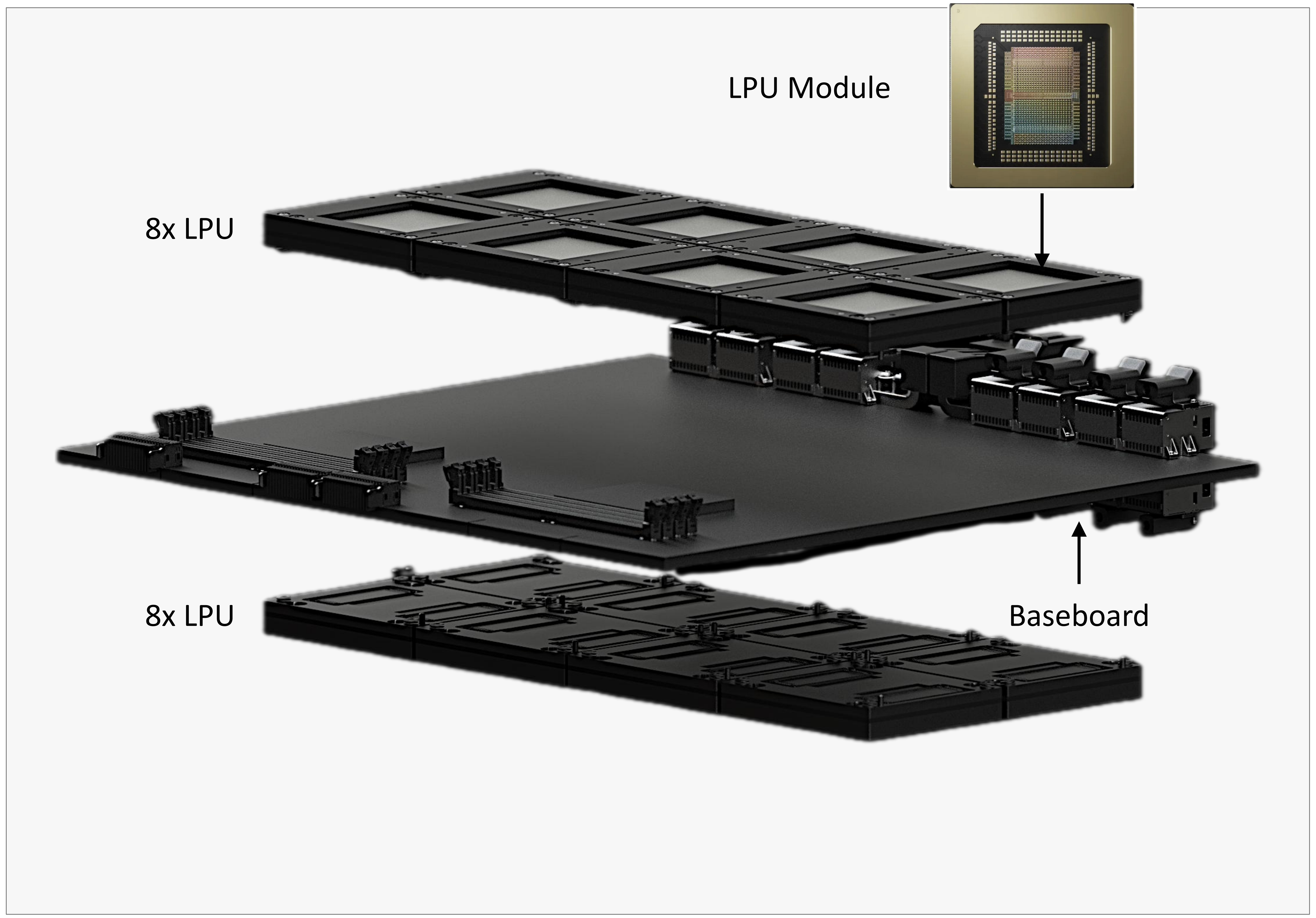
350ns Chip-to-Chip latency (from one SRAM to another SRAM)



LPX Rack
Vera Rubin Compatible | MGX Liquid Cooled
Scales to 1000+ LPUs

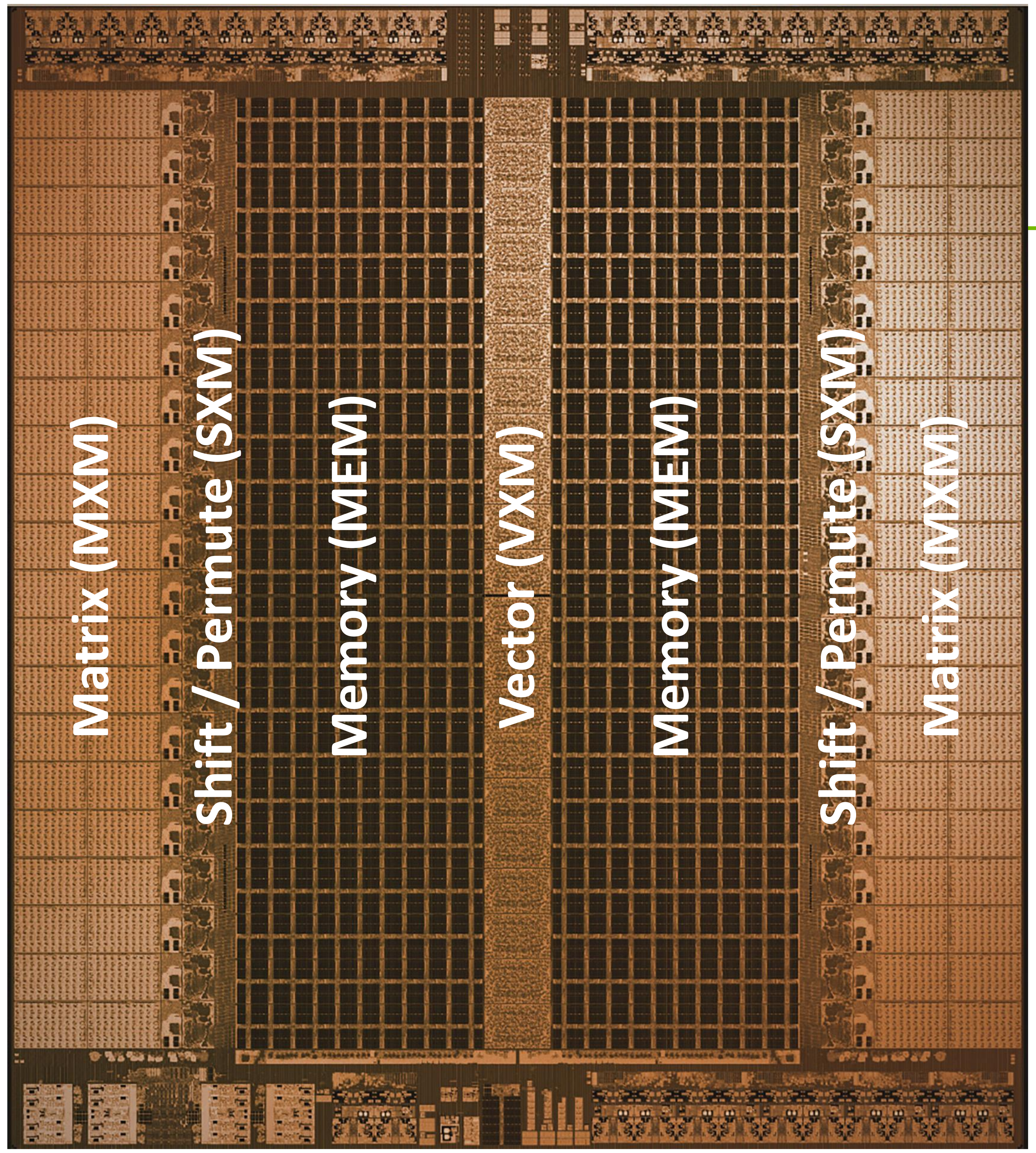
High-Reliability Design for Mass Production

NVIDIA Groq 3 LPX Rack: no tray cables, no re-timers, full-copper rack interconnect



Determinism Unlocks Continuous Hardware-Software Co-Optimization

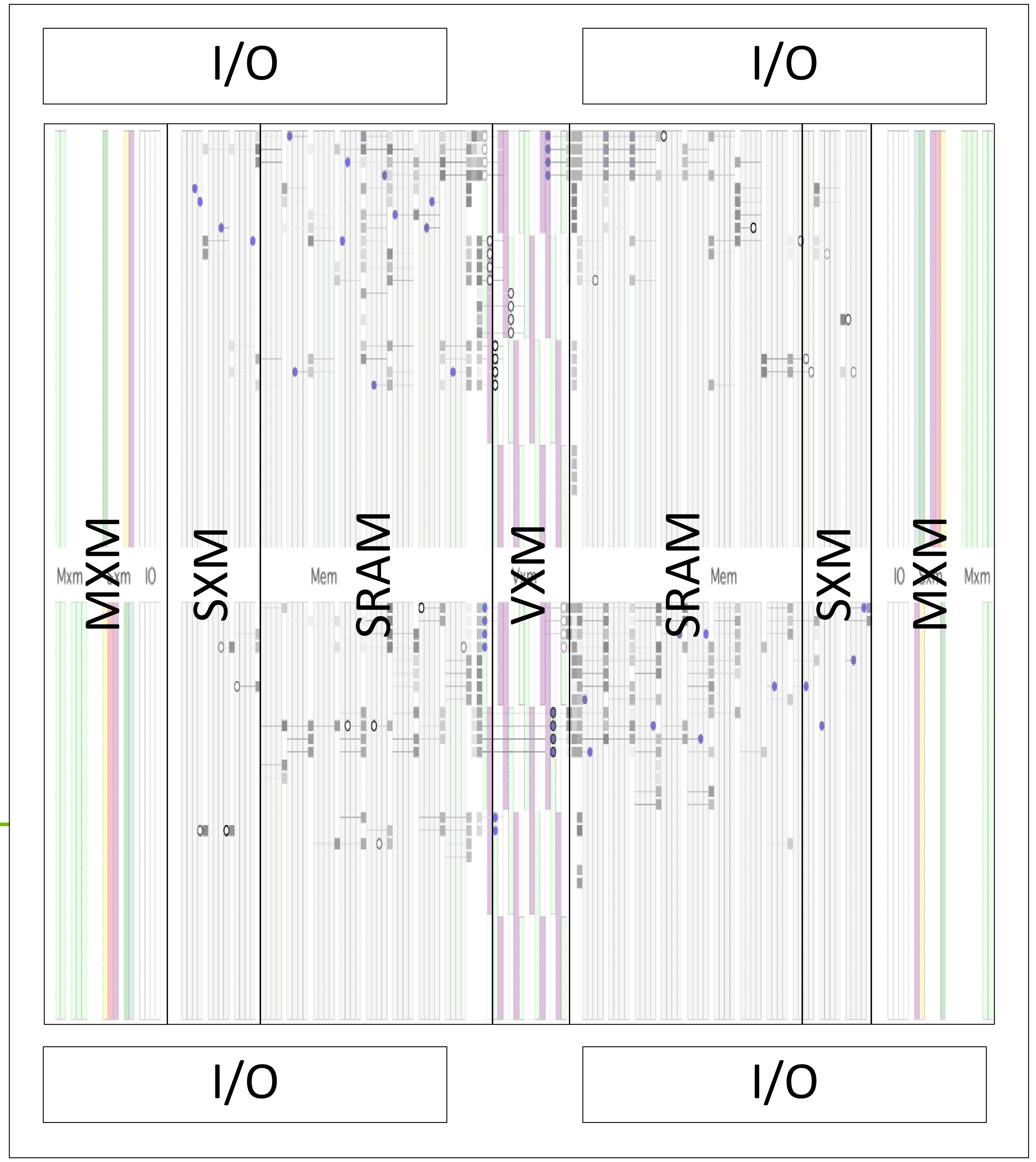
Cycle-accurate functional unit utilization allows preemptive HW environment management



Hardware and Software

Co-Optimization

2.0



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Rack-Scale Smoothing Reclaims Squandered Power

"Look-ahead adaptive suspension": send current in advance to reduce Ldi/dt, Vmin, and data center power

Determinism enables cycle-exact timetable for current demand

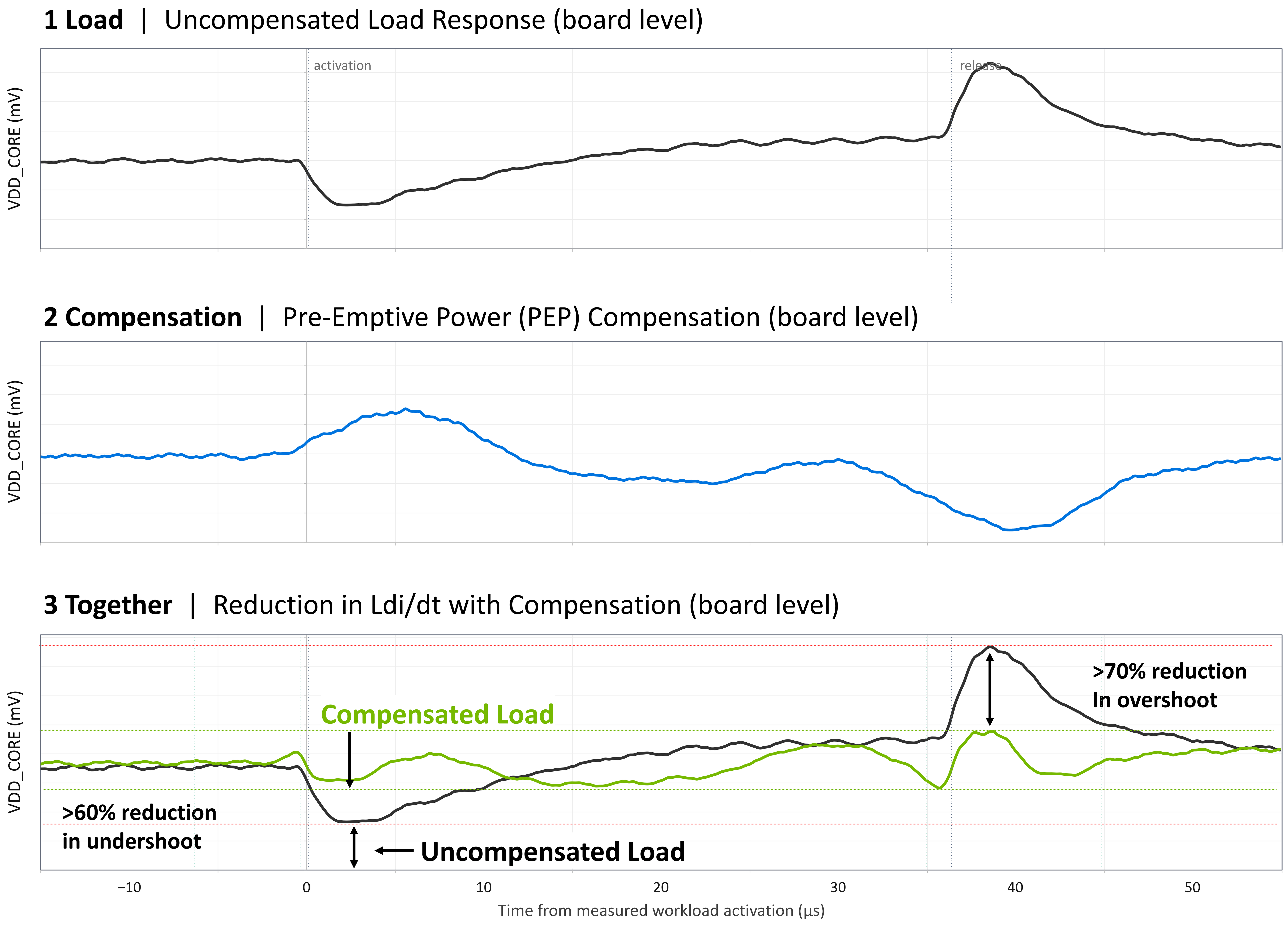
Pre-Emptive Power (PEP) orders current from regulator before it is needed

Current supply arrives at exact time as current demand reducing voltage under/overshoot

>60% less droop

>70% less overshoot

Example of Preemptive Voltage Droop Compensation



Determinism Enables Higher Thermal Design Power

Deterministic workload per-block allocation turns thermal headroom into throughput

Thermal management is a key limiter to performance

Non-deterministic processors react to the highest temperature on the chip and throttle performance

Determinism enables per-block tuning of chip utilization at compile time improving throughput

Non-Deterministic Thermal Control

Uncapped Baseline
(Tj risky, over and under utilized)

113º	128º	128º	113º
85º	124º	126º	118º

peak 128ºC · 7 blocks over limit

Uniform Power Cap
(under-utilized blocks)

94º	105º	105º	94º
74º	102º	104º	98º

hottest→105ºC · rest over-cooled

Deterministic Thermal Control

Per-Block Boost-to-Limit
(Tj risky, over and fully utilized)

105º	105º	105º	105º
105º	105º	105º	105º

every block→105ºC · reclaimed

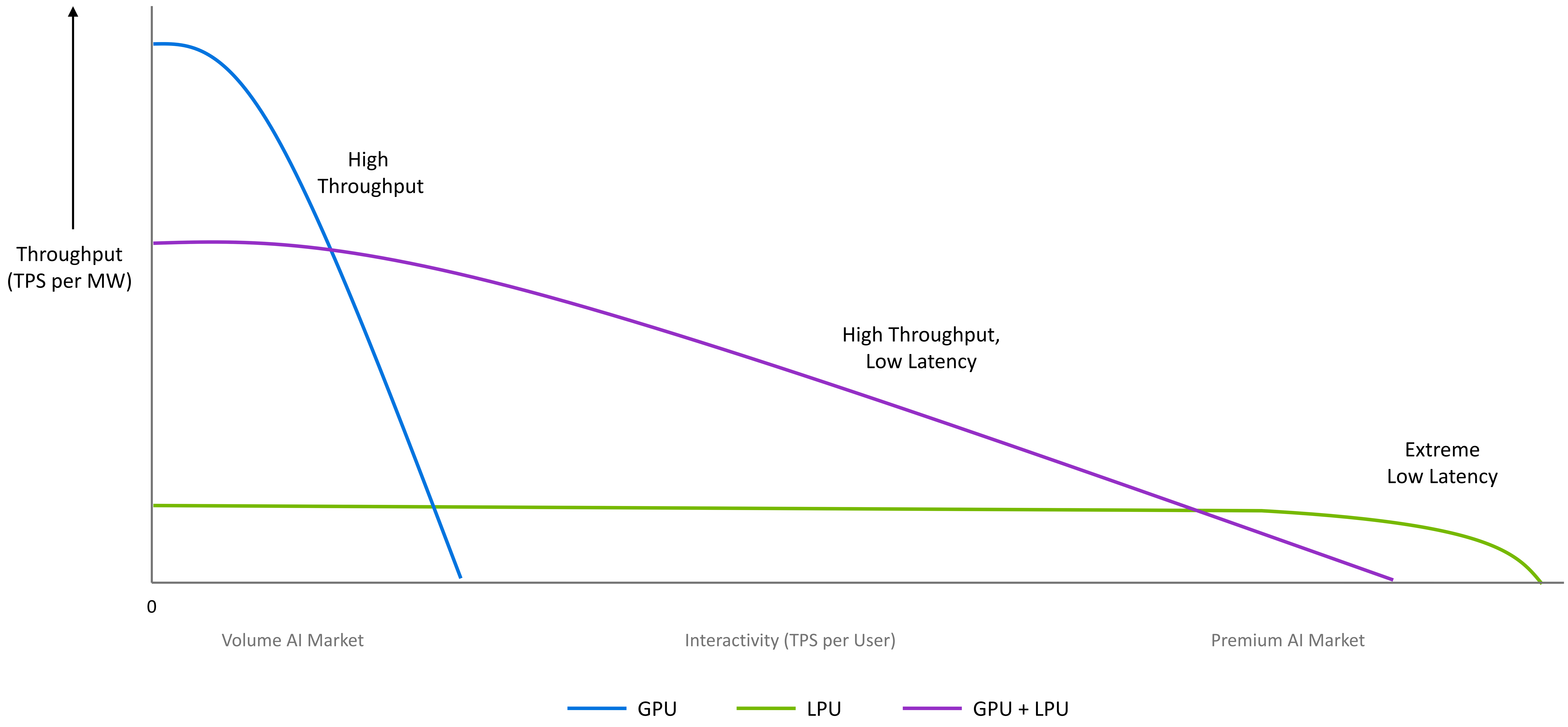
Tile color = junction temperature: **green** = headroom left · **gray** = at limit · **orange** = over limit

Heterogenous Inference With VR NVL72 + Groq 3 LPX

Santosh Raghavan

GPU-LPU Co-Design Unites Extreme Throughput With Interactive Latency

Offering full spectrum of throughput and latency



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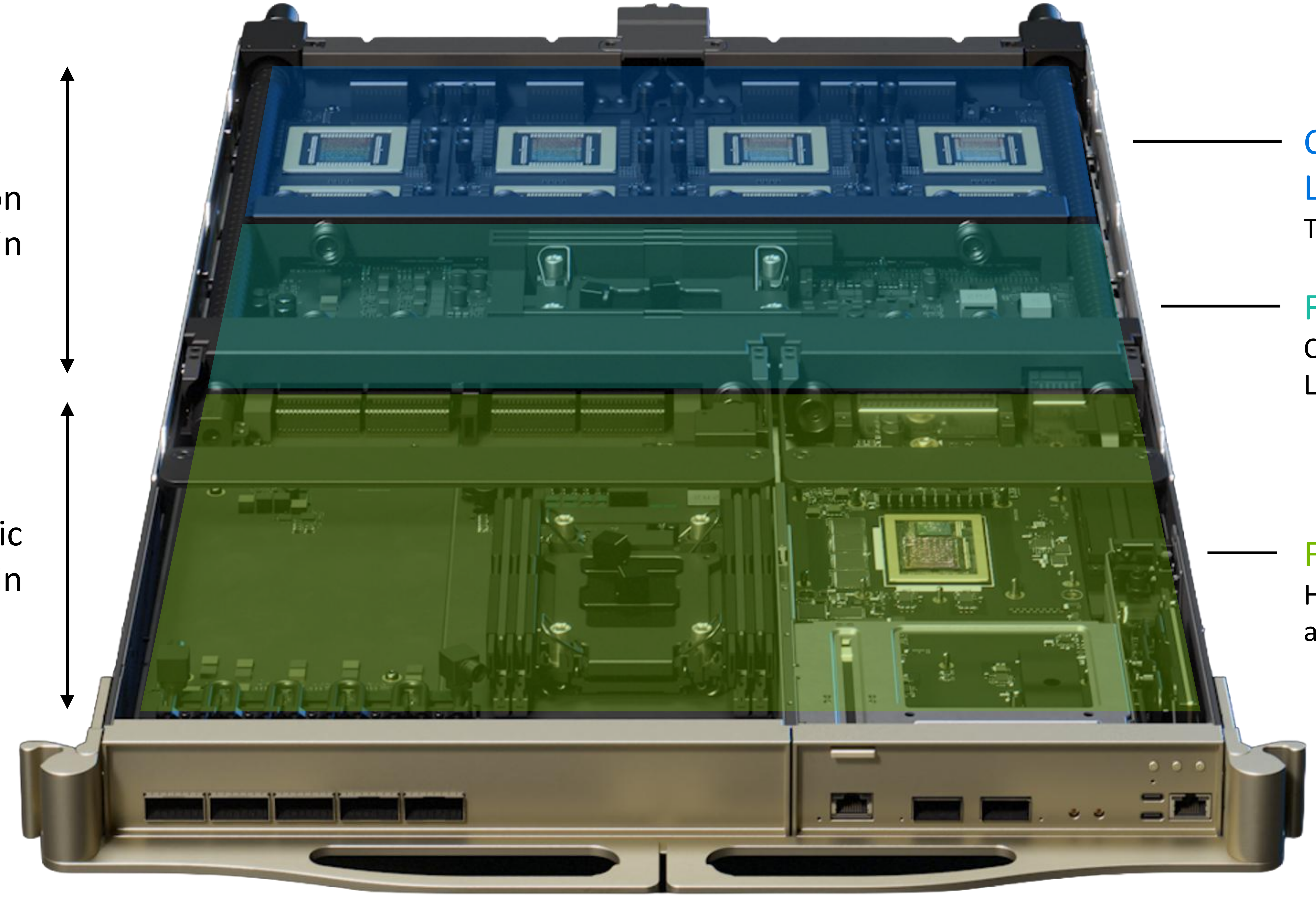
High Utilization of LPU Sync Domain Through an Async Bridge

Keep latency critical working sets in LPU / stage capacity through FPGA and HOST

LPU-LPU Network Connections



Deterministic Invocation Domain

Non-Deterministic Invocation Domain

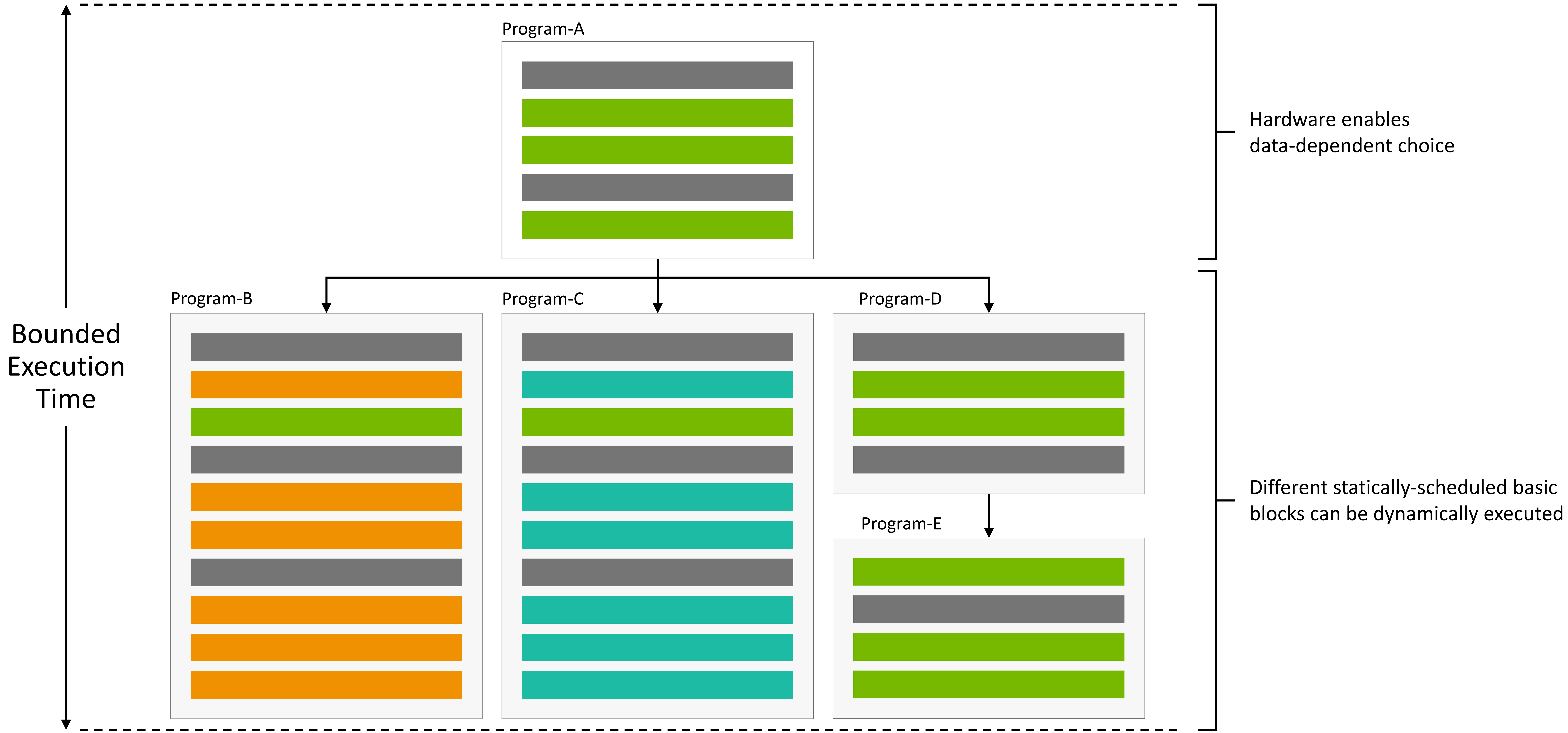
Compiler-scheduled Latency Critical
Tight cycle-accurate execution

FPGA Runtime Orchestrator
Control-plane setup, DRAM access, LPU-GPU connectivity

Flexible Host Control Plane
Host runtime, system SW to prepare, stage, and submit work

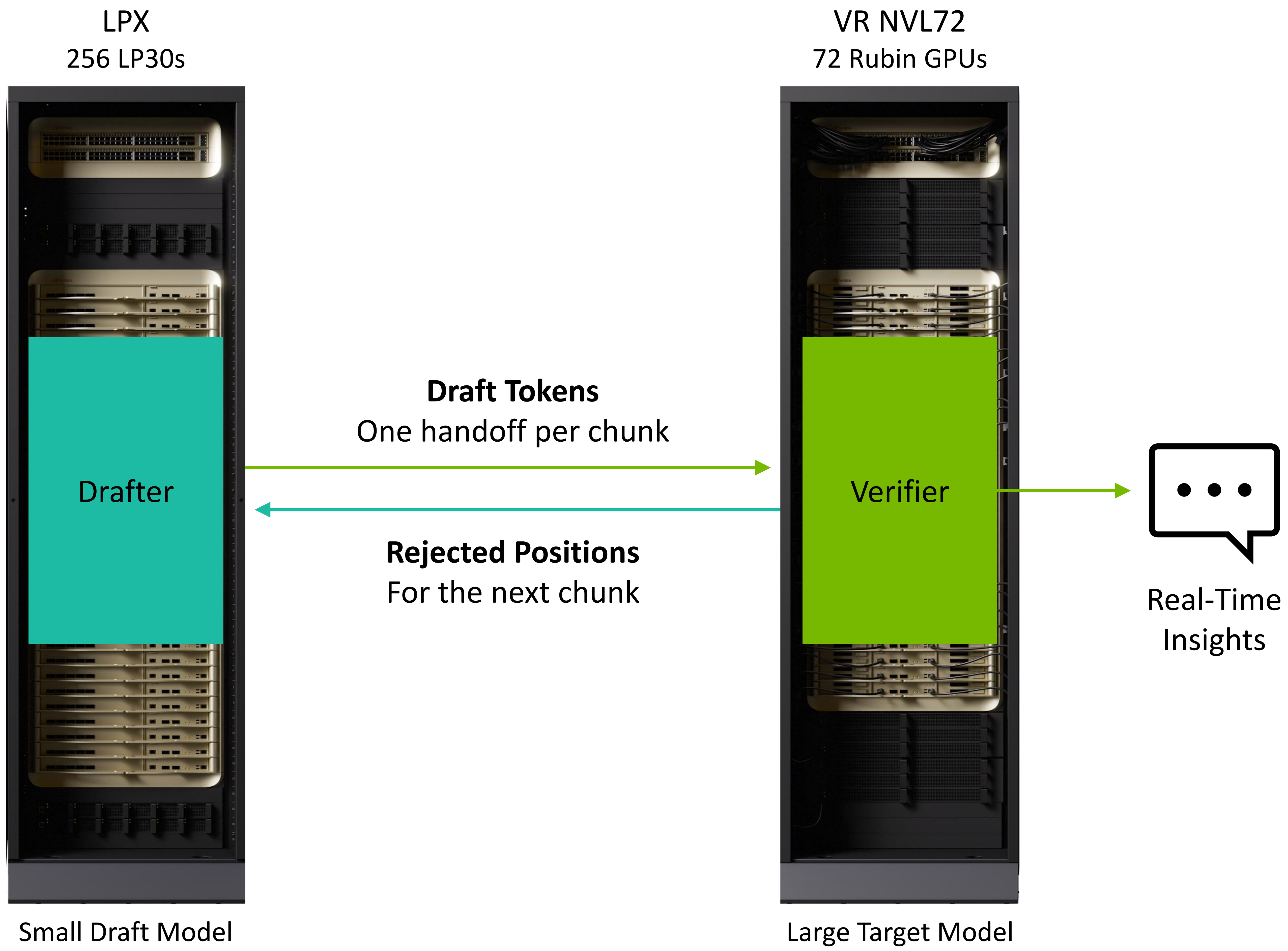
Dynamic Workloads Can Be Statically Scheduled

Branching within preset timelines



External-Drafter Speculative Decode

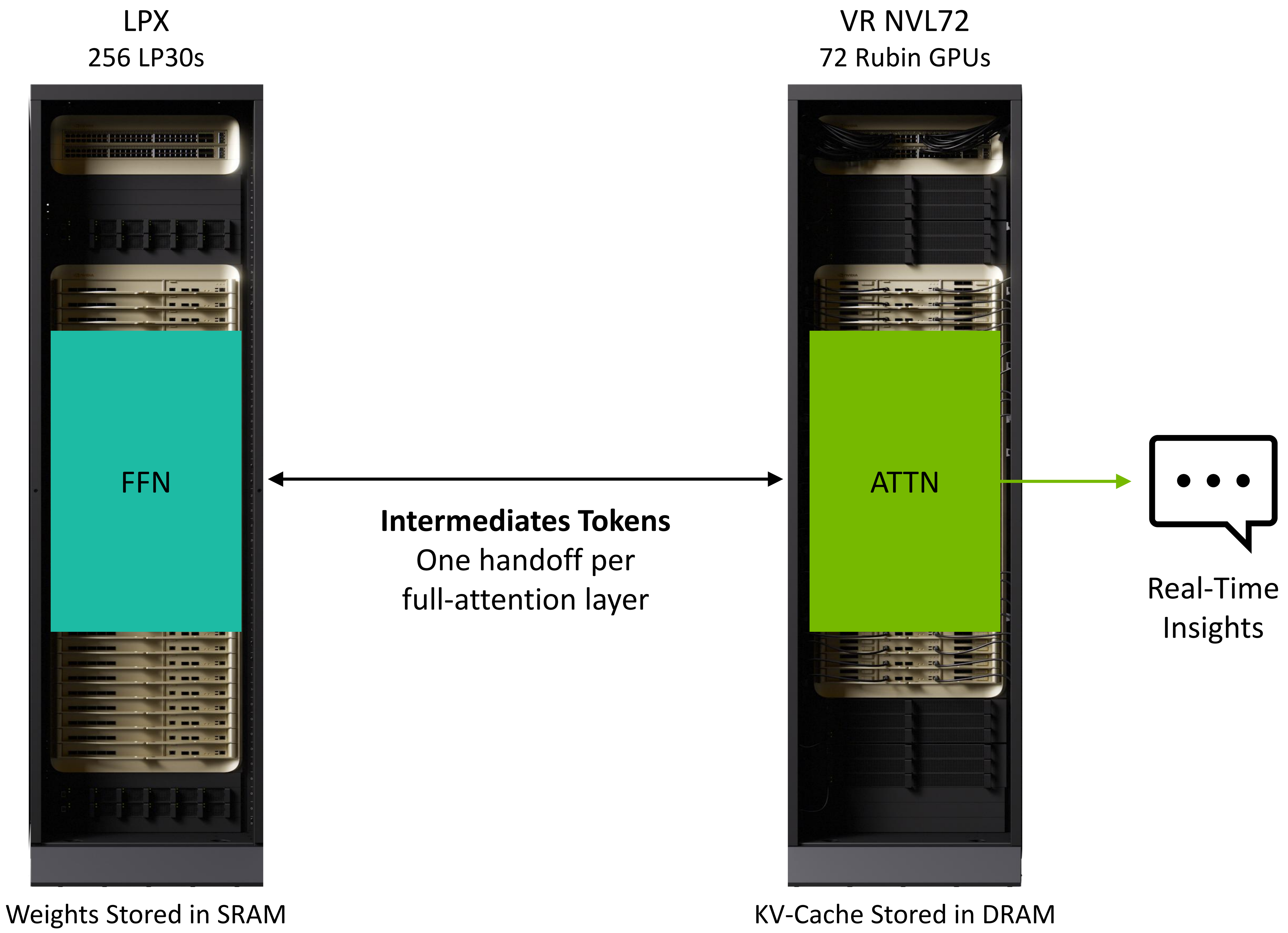
LPX drafts ahead — VR NVL72 verifies pass and commits tokens



Each rack keeps its own mode’s KV cache | **Only draft tokens cross the link**

ATTN-FFN Disaggregated Decode Across Racks

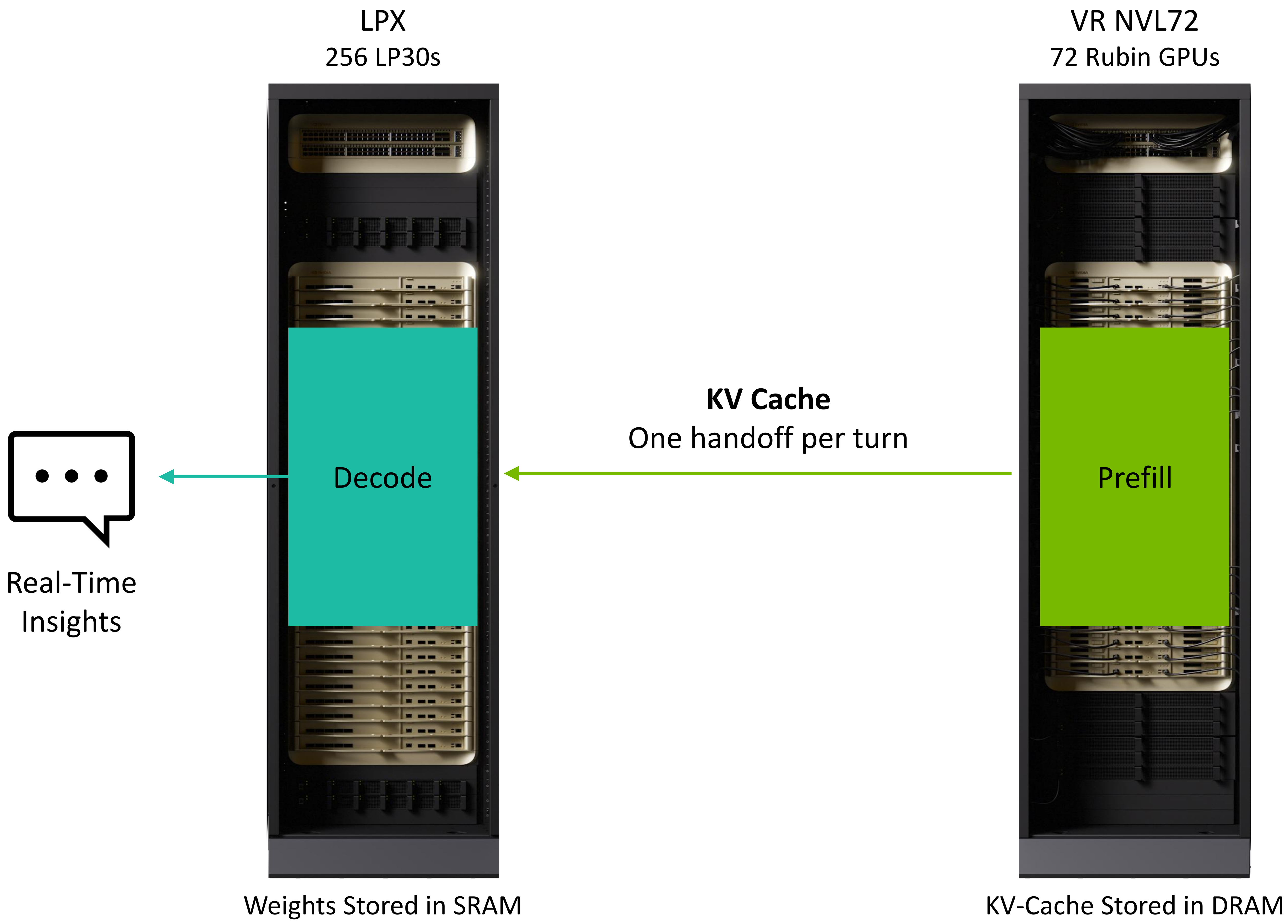
VR NVL72 computes long-context attention and stores cache — LPUs accelerate FFN layers



Allocating compute to architecture best-suited for the task

Disaggregated Prefill and Decode Maximizes Inference AI Factory Pareto Frontier

VR NVL72 computes KV cache for agentic workload — LPX delivers ultra-fast decode on long-context



Allocating compute to architecture best-suited for the task

Maximizing Utilization of GPU-LPU Pipeline

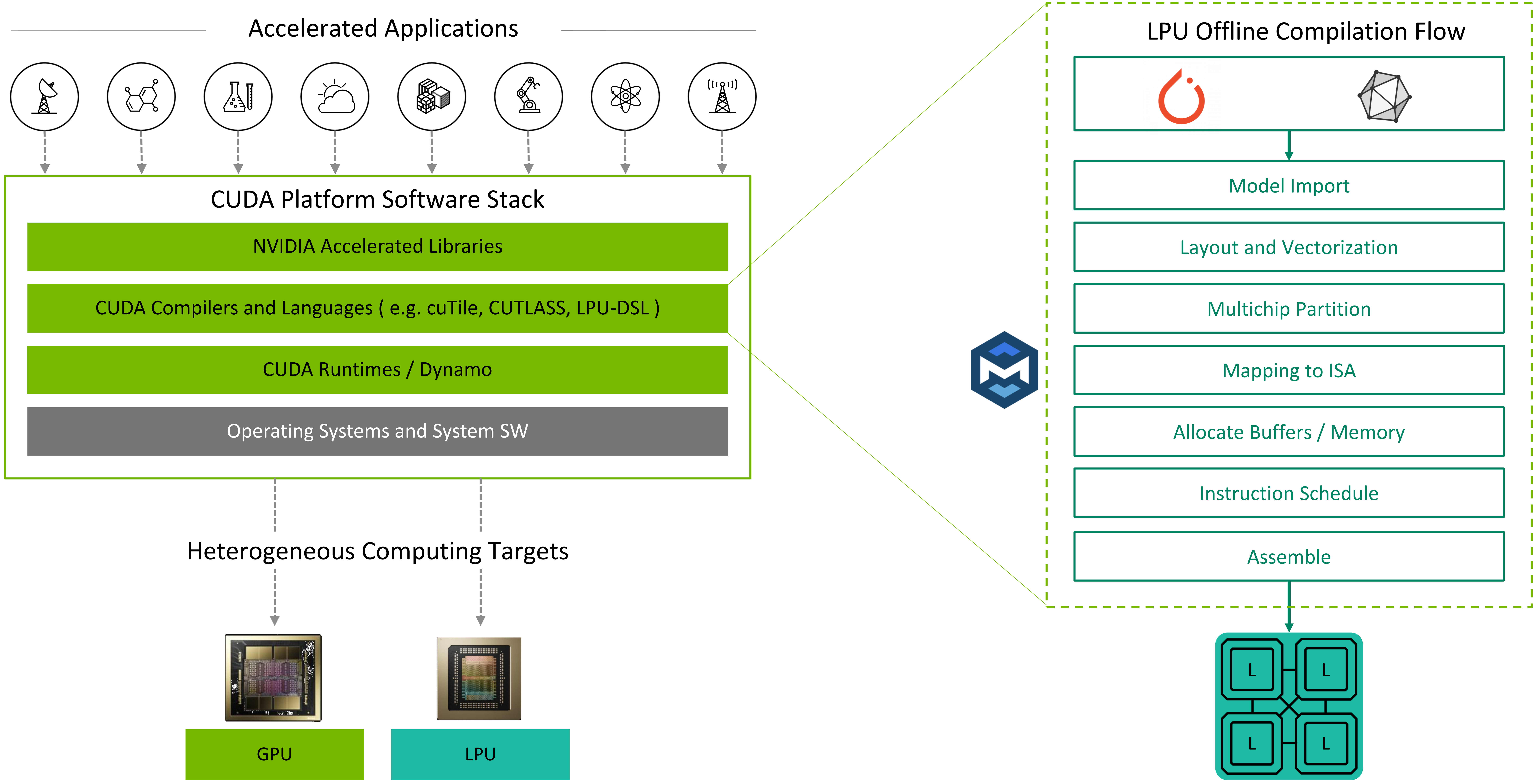
Overlapping compute and communication b/w LPUs and GPUs

LPUs **speculate low-latency** draft tokens
or
LPUs compute on FFN layers at
low-latency and high efficiency

GPUs **verify** the draft tokens at **high efficiency**
or
GPUs **compute** ATTN and **store** KV-cache
at **high efficiency**

Future CUDA Support for LPU Development

LPU as a fully-supported target by the CUDA Platform



LPU Boosts Vera Rubin Inference Throughput and Interactivity

Disaggregation benefits enhanced at large context agentic workloads

LPU Boosts Vera Rubin Inference Throughput and Interactivity

Disaggregation benefits enhanced at large context agentic workloads

GPT-OSS-2T | Cached ISL = 400K / New ISL = 4K / OSL = 400

LPU Boosts Vera Rubin Inference Throughput and Interactivity

Disaggregation benefits enhanced at large context agentic workloads

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Disaggregation benefits enhanced at large context agentic workloads

Key Takeaways

Vera Rubin co-designs every layer of the AI factory for agentic AI

Vera Rubin NVL72 is foundation platform extended by Groq 3 LPX

Deterministic HW-SW coordination unlocks power / thermal optimization

- Relaxed LPU programming model bridges determinism and workload dynamism

GPU+LPU heterogenous disaggregation brings best of Heterogeneous Compute

- Prefill-Decode Disaggregation provides highest interactivity
- Ext-drafter on LPUs improves target model verification throughput and latency on GPUs
- ATTN on GPU and FFN on LPU improves latency and utilization of decoding pipeline

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